Mth622 for final

MTH622 QUIZ#2 (lec23to27)

Date (12july2021 to 15july2021)

- Mth22 current Quiz 2 file 02-24-2021 12.16 23feb2021 (pg17)
- Mth622 Quiz2, solved by Mahar Afaq Safeer Muhammad 23feb2021 (pg28)
- Mth622 mega file quiz2 (pg233) 3files
- Mth622 Quiz2 (12-july-2021)(solution with proof) (pg17)
- Mth622 current Quiz. Made by Stylish 12july2021 (pg19)
- Mth622 quiz2 current 13july2021 (pg8)
- Mth622 current quiz2 solution-1.pdf (pg107)
- (mth622 Searching file by Skhan Academy) 14july2021 (pg107)
- Mth622 current quiz 2 Solution 14july2021 (pg106)
- Mth622 current quiz2,14july2021 made by Stylish (pg98)

MTH622:Quiz#2

Question # 8 of 10 (Start time: 03:05:24 PM, 23 February 2021)

The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus the moment of inertia due to the total mass placed at the centroid.

Select the correct option

○	true
○	false

Module 165, pg131

MTH622:Quiz#2

Question # 3 of 10 (Start time: 02:49:55 PM, 23 February 2021)

The mass of the circular disc of density ρ and radius r is

Select the correct option

○	$2\pi\rho r$
○	$\pi\rho r^2$ Module 139, pg72

MTH622:Quiz#2

Question # 5 of 10 (Start time: 02:51:43 PM, 23 February 2021)

For an uniform elliptical plate of mass M and density

ρ

with semi major axes and semi minor axes a, b respectively, we have

Select the correct option

○	$M = \frac{\rho}{\pi ab}$
○	$M = \frac{\pi ab}{\rho}$
○	$M = \rho\pi(a+b)$
○	$M = \rho\pi ab$

Module 173, pg150

MTH622:Quiz#2

Question # 1 of 10 (Start time: 09:03:47 PM, 23 February 2021)

The Moment of inertia of a hollow sphere having mass M and radius a is given by

Select the correct option

○	$I = Ma^2$
○	$I = \frac{1}{2}Ma^2$
○	$I = \frac{2}{5}Ma^2$
○	Module 155, pg106 $I = \frac{2}{3}Ma^2$

MTH622:Quiz#2

Question # 9 of 10 (Start time: 03:06:21 PM, 23 February 2021)

The mass of a cylindrical shell of density ρ, height h thickness dr and radius r is

Select the correct option

○	$2\pi\rho r dr h$
○	$\pi r^2 \rho dr h$

A

Question # 5 of 10 (Start time: 03:02:20 PM, 23 February 2021)

The Moment of inertia of a solid sphere having mass M and radius a is given by

Select the correct option

○	$I = Ma^2$
○	$I = \frac{1}{2}Ma^2$
○	Module 136 $I = \frac{2}{5}Ma^2$
○	$I = \frac{2}{3}Ma^2$

MTH622:Quiz#2

Question # 7 of 10 (Start time: 02:53:13 PM, 23 February 2021)

For uniform solid sphere

Select the correct option

○	$I_{xx} = I_{yy} = I_{zz}$
○	$I_{xx} + I_{yy} + I_{zz} = 0$

Module 154, pg102

MTH622:Quiz#2

Question # 1 of 10 (Start time: 02:57:34 PM, 23 February 2021)

The Moment of inertia of a hollow cylindrical shell having mass M and radius a is given by

Select the correct option

○	Module 142, pg81	$I = Ma^2$
○		$I = \frac{1}{2}Ma^2$
○		$I = \frac{2}{5}Ma^2$
○		$I = \frac{2}{3}Ma^2$

Question # 3 of 10 (Start time: 03:00:19 PM, 23 February 2021)

Three particles of masses m, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find I_{yy}

Select the correct option

○	7m
○	8m

Module 160, pg120

MTH622:Quiz#2

Question # 7 of 10 (Start time: 03:04:23 PM, 23 February 2021)

Three particles of masses m, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find I_{zz}

Select the correct option

- ○ 9m

Module 160,
pg120

- ○ 8m

MTH622 Quiz#2

Question # 4 of 10 (Start time: 02:50:38 PM, 23 February 2021)

The Moment of inertia of a solid cylinder having mass M and radius a is given by

Select the correct option

○	$I = Ma^2$
○	$I = \frac{1}{2}Ma^2$
○	$I = \frac{2}{5}Ma^2$
○	$I = \frac{2}{3}Ma^2$

Module 152, pg99

MTH622:Quiz#2

Question # 2 of 10 (Start time: 02:59:18 PM, 23 February 2021)

The equation for an elliptical plate with semi major axes along x-axis is given by

Select the correct option

○	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
○	$\frac{x^2}{b^2} - \frac{y^2}{a^2} = 1$
○	Module 151, pg96 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
○	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

MTH622:Quiz#2

Question # 10 of 10 (Start time: 03:08:15 PM, 23 February 2021)

Three particles of masses m, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find I_{xx}

Select the correct option

- ○ 7m

  Module 120, pg160

- ○ 8m

MTH622:Quiz 2

Question # 1 of 10 (Start time: 11:35:04 AM, 15 July 2021)

The force field $\vec{F} = -x\hat{i} - 2z\hat{k}$ is

Select the correct option

○	non conservative
○	conservative

B

Question # 9 of 10 (Start time: 01:47:41 PM, 14 July 2021)

Total

Simple Harmonic Motion is a particular type of oscillation and periodic motion in which restoring force of an object is .... to the displacement of the object acting in opposite direction of displacement.

Select the correct option

○	Directly proportional
○	Inversely proportional

Module 102, pg11

MTH622:Quiz 2

Question # 2 of 10 (Start time: 09:23:39 AM, 15 July 2021)

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{3}{4}t)$. Then the amplitude is

Select the correct option

●	20 Module 104, pg15
○	$\frac{3}{4}$

MTH622:Quiz 2

Question # 10 of 10 (Start time: 12:10:25 PM, 15 July 2021)

An object in rectilinear motion moves

Select the correct option

- ○ along straight line Module 111, pg25
- ○ none of these
- ○ along curved path
- ○ oscillatory

Question # 8 of 10 (Start time: 11:41:46 AM, 15 July 2021) Total Marks: 1

Law of conservation of energies can be used to derive the condition of conservation of forces.

Select the correct option

- false
- true

Module 99, pg41

MTH622:Quiz 2 — Quiz Start Time: 11:35 AM, 15 July 2021

Question # 7 of 10 (Start time: 11:41:02 AM, 15 July 2021) — Total Marks: 1

The forces acting on a damped harmonic oscillator tend to .... the amplitude of the successive oscillations.

Select the correct option

○	decrease — Module 106, pg18 ✓
○	increase

Question # 4 of 10 (Start time: 11:38:20 AM, 15 July 2021)

Total Marks: 1

The maximum distance covered by the oscillating body in one oscillation or length of a wave measured from its mean position is called amplitude.

Select the correct option

○	false
○	true

Module 102, pg11

MTH622:Quiz 2

Question # 2 of 10 (Start time: 11:36:07 AM, 15 July 2021)

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{2}{3}t)$. Then the time period is

Select the correct option

- 3π

  Module 104, pg15

- $\frac{3}{2}\pi$

MTH622:Quiz 2

Question # 6 of 10 (Start time: 10:57:34 AM, 15 July 2021)

The basis vectors for cylindrical coordinate system are

Select the correct option

- ○ $e_\rho = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$, $e_\phi = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$, and $e_z = \hat{k}$

  Module 113, pg28

- ○ $e_\phi = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$, $e_\rho = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$, and $e_z = \hat{k}$

Question # 4 of 10 (Start time: 10:55:04 AM, 15 July 2021)

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{3}{4}t)$. Then the time period will be

Select the correct option

- $\frac{8}{3}\pi$

Module 104, pg15

- $\frac{4}{3}\pi$

Question # 3 of 10 (Start time: 10:54:08 AM, 15 July 2021) Total Marks: 1

Euler's theorem states that rotation of a rigid body about a fixed point of the body is equivalent to a .... about a line which passes through the point.

Select the correct option

- displacement
- reflection
- rotation Module 108, pg21
- translation

Question # 2 of 10 (Start time: 02:04:30 PM, 15 July 2021) Total Marks: 1

Time period is .... required by a oscillation system to complete its one cycle of oscillation of the specific system.

Select the correct option

- minimum time

  Module 103, pg13

- maximum time

MTH622:Quiz 2

Question # 4 of 10 (Start time: 08:34:10 PM, 15 July 2021)

A rigid body has....... degrees of freedom,

Select the correct option

- six — Module 109, pg23 ✓
- three
- infinite many
- four

Question # 1 of 10 (Start time: 04:01:08 PM, 15 July 2021)

Total Marks: 1

Forces that cannot be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- false
- true

Module 100, pg43

Question # 10 of 10 (Start time: 08:03:03 PM, 15 July 2021)

A freely falling object observes

Select the correct option

- uniform rectilinear motion
- none of these
- oscillatory motion
- uniformly accelerated motion

Module 111, pg26

MTH622:Quiz 2

Question # 7 of 10 (Start time: 08:00:15 PM, 15 July 2021)

If position of a vibrating mass at any time is defined as $y(t) = 10sin(\frac{1}{3}t)$. Then the amplitude is

Select the correct option

○	10 Module 104, pg15 ✓
○	$\frac{1}{3}$

MTH622:Quiz 2

Question # 5 of 10 (Start time: 07:58:00 PM, 15 July 2021)

The force field $\vec{F} = 3x\hat{i} - 2y\hat{j}$ is

Select the correct option

○	conservative Module 98, pg36
○	non conservative

Question # 5 of 10 (Start time: 08:34:42 PM, 15 July 2021) Total Marks: 1

The restoring force of a harmonic oscillator is proportional to ......

Select the correct option

- acceleration
- time
- displacement — Module 102, pg11
- velocity

Question # 6 of 10 (Start time: 08:35:20 PM, 15 July 2021) Total Marks: 1

The trajectory of a particle in projectile motion with air resistance is

Select the correct option

○	none of these
○	a parabola
○	a non parabolic curve Module 114, pg31
○	a hyperbola

MTH622:Quiz 2 Quiz Start Time: 09:28 PM, 15 July 2021

Question # 3 of 10 (Start time: 09:30:04 PM, 15 July 2021) Total Marks: 1

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line followed by a .... about that line.

Select the correct option

- ○ translation
- ● rotation — Module 109, pg43
- ○ reflection
- ○ displacement

Question # 8 of 10 (Start time: 09:35:18 PM, 15 July 2021) Total Marks: 1

The damping force of a harmonic oscillator is proportional to its ......

Select the correct option

○	acceleration
○	time
○	displacement
●	velocity Module 106, pg18 ✓

Question # 2 of 10 (Start time: 09:39:45 PM, 15 July 2021)

Total Marks: 1

Uniformly accelerated rectilinear motion is a special case of
uniform rectilinear motion.

Select the correct option

○	false Module 111, pg26
○	true

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 09:42:28 PM, 15 July 2021) Total Marks: 1

Kinematics is the branch of mechanics that deals with the moving
objects and the forces which cause the motion.

Select the correct option

- ○ false — Module 110, pg24
- ○ true

Question # 10 of 10 (Start time: 09:37:46 PM, 15 July 2021)

A car driving along a curved road is an example of

Select the correct option

○	oscillatory motion
○	curvilinear motion — Module 112, pg27 ✓
○	none of these options
○	rectilinear motion

MTH622:Quiz 2 Quiz Start Time: 09:49 PM, 15 July 2021

Question # 1 of 10 (Start time: 09:49:58 PM, 15 July 2021) Total Marks: 1

Which of the following are examples of simple harmonic motion

Select the correct option

○	both (a) and (b) options Module 102, pg11_12
○	(c) movement on a slide
○	(b) object of mass m linked to a spring
○	(a) movement of the mass attached to a simple pendulum

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 11:03:59 PM, 15 July 2021)

Elastic spring force is not an example of conservative force.

Select the correct option

○	false Module 97, Pg39 ✓
○	true

Question # 2 of 10 (Start time: 11:04:48 PM, 15 July 2021)

An object of mass m linked to a spring of spring constant k represents the

Select the correct option

○	simple harmonic motion Module 102, pg11
○	rectilinear motion

Question # 4 of 10 (Start time: 11:06:27 PM, 15 July 2021)

In case of conservative force field, the total energy is........

Select the correct option

- a constant — Module 99, pg41 ✓
- any function of time

MTH622:Quiz 2

Question # 6 of 10 (Start time: 11:08:01 PM, 15 July 2021)

The force field $\vec{F} = y\hat{j} + z\hat{k}$ is

Select the correct option

○	conservative A
○	non conservative

MTH622:Quiz 2

Question # 10 of 10 (Start time: 11:10:41 PM, 15 July 2021)

In projectile motion, the air resistance is always neglected.

Select the correct option

○	true Module 114, pg30 ✓
○	false

MTH622:Quiz 2

Question # 1 of 10 (Start time: 11:21:41 PM, 15 July 2021)

For the conservative force field $\vec{F} = -2\hat{i}x$ the scalar potential V will be

Select the correct option

○	$x^2 + c$ A
○	$xy + c$

MTH622:Quiz 2

Question # 2 of 10 (Start time: 11:22:30 PM, 15 July 2021)

Gravitational force is an example of a conservative force.

Select the correct option

○	true Module 97, Pg39
○	false

MTH622:Quiz 2

Question # 3 of 10 (Start time: 11:23:08 PM, 15 July 2021)

For the conservative force field $\vec{F} = -2\hat{j}y$ the scalar potential V will be

Select the correct option

○	$y^2 + c$ A
○	$xy + c$

MTH622:Quiz 2

Question # 5 of 10 (Start time: 11:24:31 PM, 15 July 2021)

Friction is a

Select the correct option

- conservative force
- non conservative force

Module 100,
pg43

MTH622:Quiz 2

Question # 6 of 10 (Start time: 11:25:09 PM, 15 July 2021)

Find the time period T of the vibrating mass with $m = 10$ and $f = 30$

Select the correct option

○	$T = \frac{1}{3}$
○	$T = \frac{1}{30}$ Module 103, pg13 ✓

MTH622:Quiz 2

Question # 7 of 10 (Start time: 11:25:41 PM, 15 July 2021)

Find the time period T of the vibrating mass with $m = 5$ and $f = 10$

Select the correct option

- $T = \frac{1}{2}$
- $T = \frac{1}{10}$

Module 103, pg13

Question # 9 of 10 (Start time: 09:57:17 PM, 15 July 2021) Total Marks: 1

Which of the following force is not conservative

Select the correct option

- ● frictional force — Module 97, pg39
- ○ electric force
- ○ elastic spring force
- ○ gravitational force

Question # 1 of 10 (Start time: 01:38:27 PM, 14 July 2021)

The frequency f of an oscillatory system is equal to

Select the correct option

- $f = e^{-T}$
- $f = e^{T}$
- $f = -\frac{1}{T}$
- $f = \frac{1}{T}$

Module 103, pg 13

MTH622:Quiz 2 Quiz Start

Question # 5 of 10 (Start time: 05:59:00 PM, 14 July 2021)

A continuously differentiable force field $\vec{F}$ is conservative if and only if

Select the correct option

- (a) for any closed non- intersecting curve C (simple closed curve) $W = \oint_C \vec{F}.d\vec{r} = 0.$

Module 97, pg38_39

- $(c)\vec{F} = -\nabla\varphi$ for a continuously differentiable scalar field φ.
- all a,b,c options are true
- $(b)curl\vec{F} = \nabla \times \vec{F} = 0.$

Question # 8 of 10 (Start time: 10:36:40 AM, 12 July 2021)

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

●	false Module 100, pg43
○	true

Mth622 QUIZ#3

Lec (28-34) Topic (119-145)

Date (2 August 2021 - 6 August 2021)

- **Mth622 Quiz NO3.pdf Skhan Academy (pg79)**
- **Mth622 current quiz 5August.pdf (pg31)**
- **Cam scanner 08-05-2021 (pg11)**
- **Cam scanner 08-05-2021 8.26.pdf (pg19)**

Question # 9 of 10 (Start time: 01:01:59 PM, 05 August 2021) **Total Marks:**

The torque acting on a particle is proportional to the time rate of change in its angular acceleration.

Select the correct option

●	false
	true

Question # 1 of 10 (Start time: 11:50:55 AM, 05 August 2021)

The product of inertia of a particle about x and y-axes is

Select the correct option

○	$I_{xy} = m(x^2 + y^2)$
○	$I_{xy} = -mxy$

B

Question # 8 of 10 (Start time: 11:48:12 AM, 05 August 2021) Total Marks: 1

In general, the directions of angular velocity and angular momentum are

Select the correct option

- ○ distinct

  A

- ○ same

Question # 7 of 10 (Start time: 01:00:20 PM, 05 August 2021) Total Marks: 1

If P is any point on the ellipsoid of inertia, then the moment of inertia of a rigid body about any line $\overrightarrow{OP}$ is equals to the reciprocal of the square of the length of $|\overrightarrow{OP}|$.

Select the correct option

Reload Math Equations

- false
- true

B

Question # 8 of 10 (Start time: 11:55:29 AM, 05 August 2021) Total Marks

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow ......

Select the correct option

- a straight line
- a parabola

B

TH622 Quiz 3

Quiz Start Time: 11:50 AM, 05 August 20

uestion # 6 of 10 (Start time: 11:54:19 AM, 05 August 2021)

Total Mark

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$, find m

elect the correct option

Reload Math Equatio

○	4
○	3

n # 5 of 10 (Start time: 11:53:33 AM, 05 August 2021)   Total

is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}\,a$

ne correct option

Reload Math Eq

$I = \frac{2}{5}Ma^2$

A

$I = \frac{5M}{2a^2}$

Question # 3 of 10 (Start time: 11:45:09 AM, 05 August 2021)

The moment of inertia of a particle about y-axis is

Select the correct option

- $I_{yy} = mxz$
- $I_{yy} = m(x^2 + y^2)$

B

MTH622:Quiz 3 Quiz Start Time: 11:43 AM, 05 August 2021

Question # 2 of 10 (Start time: 11:44:35 AM, 05 August 2021) Total Marks: 1

Dynamics deals with the forces acting on any object and relationship fundamentally to the motion.

Select the correct option

- true

A

- false

Question # 5 of 10 (Start time: 11:46:23 AM, 05 August 2021)

Angular momentum can be defined as

Select the correct option

- $L = r \times v$
- $L = p \times \omega$
- $L = r \times \omega$
- $L = r \times p$

D

MTH622:Quiz 3

Question # 6 of 10 (Start time: 11:47:02 AM, 05 August 2021)

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

○	$T = \frac{1}{2} I_0 \omega^2$
○	$T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

B

Question # 10 of 10 (Start time: 11:49:33 AM, 05 August 2021)

Total Marks: 1

For an elastic body, the distance between two particles.........under
the action of applied force

Select the correct option

○	remains constant
○	is changed

B

Question # 9 of 10 (Start time: 11:48:48 AM, 05 August 2021)

The radius of gyration K of the hemisphere of mass M and moment of inertia $I = \frac{2}{5}Ma^2$ about an axis through its center is

Select the correct option

○	$K = \frac{1}{a}\sqrt{\frac{5}{2}}$
○	$K = \sqrt{\frac{2}{5}}a$

B

MTH622:Quiz 3

Question # 5 of 10 (Start time: 12:58:58 PM, 05 August 2021)

If angular momentum $\Omega = 2\omega \sin \omega t$ then the torque τ is

Select the correct option

- $2\omega^2 \cos \omega t$ Ans

A

- $-2 \cos \omega t$

Question # 4 of 10 (Start time: 12:57:36 PM, 05 August 2021) **Total Marks: 1**

Consider a right circular cone of density ρ, radius a and height h is composed of elementary circular discs of small thickness each parallel to the base of the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

Reload Math Equations

○	$\delta m = \rho 2\pi r \delta z$
○	$\delta m = \rho \pi r^2 \delta z$

B

Question # 3 of 10 (Start time: 12:56:21 PM, 05 August 2021)

If a body rotates about some external fixed point it is called revolution

Select the correct option

- revolution

A

- spin

For inertia matrix, along principal axes

Select the correct option

- ○ the diagonal elements are zero
- ○ the off diagonal elements are zero

B

Question # 4 of 10 (Start time: 11:45:47 AM, 05 August 2021)

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.......

Select the correct option

○	coincident 
○	perpendicular

Question # 1 of 10 (Start time: 11:43:47 AM, 05 August 2021)

Total Marks: 1

If a particle of mass 4 moving with velocity $4\vec{i}$ collides another particle of mass 5 with velocity $3\vec{i}$. Then what will be the velocity of first particle after collision if the velocity of second particle after collision is $2\vec{i}$.

Select the correct option

Reload Math Equations

- $\frac{15}{4}\vec{i}$
- $\frac{21}{4}\vec{i}$

B

Question # 7 of 10 (Start time: 08:44:03 PM, 04 August 2021)

The position vector of the center of mass of a system of particles is

Select the correct option

$$\vec{r}_c = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$

A

$$\vec{r}_c = \frac{\sum_i m_i \vec{r}_i}{\sum_i \vec{r}_i}$$

Question # 6 of 10 (Start time: 09:54:50 AM, 05 August 2021)

If a rigid body is rotating about a fixed point on ground then its kinetic energy T is defined by

Select the correct option

○	$T = \frac{1}{2} I_0 \omega^2$
○	$T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

A

Question # 5 of 10 (Start time: 09:54:13 AM, 05 August 2021)

If the axis of rotation passes through
the center of mass of the rigid body then it is called

Select the correct option

○	spin
○	revolution

A

Question # 10 of 10 (Start time: 02:57:41 PM, 02 August

If an object explodes, the center of mass will

Select the correct option

- ○ come into rest
- ● continue its previous motion

B

Mth622 for final

- Mth622 Quiz#2 All in 1 file (pg54)
- Mth622 quiz#3 All in 1 file (pg24)
- Mth622 mcq's pdf pg187
- Mth622 mega file final pg215

  1- Mth622 by waseem Gill (pg22)

  2-Mth622 notes converted pg10
- Mth622 mega file quiz2 (final start 27-51,158-205) pg233
- Mth622-1 (pg12)
- Mth622-mid final- subjective unsolved (pg14)
- Mth622 important topic for final term (pg32)

MTH622 Vectors and Classical Mechanics

Question No : 23 of 41

The path of a projectile in a non-resistive medium is

Answer (Please select your correct option)

linear

Circular

Parabolic

Hyperbolic

MTH622 Vectors and Classical Mechanics

Question No : 21 of 41

The rate of change of displacement of a body is called

Answer (Please select your correct option)

- Velocity
- Acceleration
- Momentum
- None of these

MTH622 Vectors and Classical Mechanics

Question No :

Work done is said to be zero, when

Answer (Please select your correct option)

a. Some force acts on a body, but displacement is zero

b. No force acts on a body but some displacement takes place

Either a) or b)

None of these

MTH622 Vectors and Classical Mechanics

Question No : 25 of 41

The path of projectile is called its

Answer (Please select your correct option)

- velocity
- acceleration
- trajectory
- none of these

Question No : 30 of 41

Measure of average kinetic energy of molecules is

Answer (Please select your correct option)

- temperature
- energy
- internal energy
- enthalpy

Question No : 29 of 41

If there is no net force acting on body, then its acceleration is

Answer (Please select your correct option)

- zero
- constant
- increasing
- decreasing

MTH622 Vectors and Classical Mechanics

Question No : [illegible]

Marks: 1 (Budgeted Time [illegible])

A 3×3 symmetrical matrix has ------------- real eigenvalues, which may be distinct or repeated

Answer (Please select your correct option)

- Infinite
- Zero
- Two
- Three

Mod 175, Pg 155

Scanned with CamScanner

Question No : 7 of 41

The axes relative to which product of inertia are zero are called

Answer (Please select your correct option)

- principal axes

Mod 128, Pg 50

- principal moment of inertia
- moment of inertia
- axes of rotation

Start Time 10:56 AM

116:00

7

Question No : 5 of 41 Marks: 1 (Budgeted Time 1 Min)

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

Answer (Please select your correct option)

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$$

Mod 178

$$\left(\frac{d\vec{A}}{dt}\right)_f = A_r - \omega \times \vec{A}_r$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - A \times \vec{A}_r$$

Start Time 10:56 AM

16:00

6

Question No : 2 of 41 Marks: 1 (Budgeted Time

Which of the following property is not true?

Answer (Please select your correct option)

- $\nabla.(\vec{A}+\vec{B}) = \nabla.\vec{A} + \nabla.\vec{B}$
- $\nabla.(\varphi\vec{A}) = \varphi(\nabla.\vec{A}) + \vec{A}.(\nabla\varphi)$
- $\nabla.\nabla\varphi = \nabla^2\varphi$
- $\nabla.\vec{V} = \vec{V}.\nabla$

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MTH622 Vectors and Classical Mechanics

Question No : 9 of 41

The polar moment of inertia of a circular section is about

Answer (Please select your correct option)

- X-X axis
- Y-Y axis
- Z-Z axis
- Neutral axis

MTH622 Vectors and Classical Mechanics

Question No : [illegible]

The wheels of a moving car possess

Answer (Please select your correct option)

- Potential energy only
- Kinetic energy of translation only
- Kinetic energy of rotation only
- Kinetic energy of translation and rotation both

Question No : [illegible]

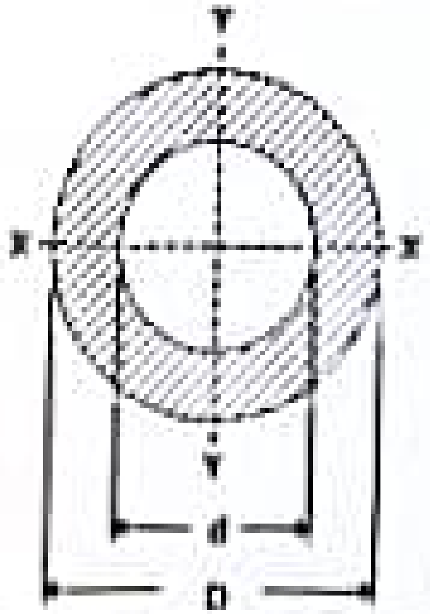
Rotational kinetic energy of disc is given by

Answer (Please select your correct option)

- $1/4(mv^2)$
- $1/2(mv^2)$
- $1/8(mv^2)$
- $1/6(mv^2)$

Module 141, pg 78

Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is



swer (Please select your correct option)

$\pi/16\ (D^4 - d^{[illegible]})$

$\pi/16\ (D^3 - d^{[illegible]})$

$\pi/16\ (D^4 - d^{[illegible]})$

$\pi/64\ (D^4 - d^4)$

MTH622 Vectors and Classical Mechanics

Question No : 17 of 45

At pivot point, body with mass 'm' will have a torque by th

Answer (Please select your correct option)

- force
- circle
- central point
- radius of circle

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Question No : [illegible] Marks: 1 Unassigned Time [illegible]

The periodic time of a particle with simple harmonic motion is __________ proportional to the angular velocity

Answer (Please select your correct option)

- Directly
- Inversely
- Square root
- None of these

110:00
Time Left

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MTH622 Vectors and Classical Mechanics

Question No : 22 of 41

For an irrotational flow

Answer (Please select your correct option)

- $\nabla \times \vec{v} = \vec{0}$
- $\nabla \times \vec{v} \neq \vec{0}$
- $\nabla . \vec{v} = \vec{0}$
- None of these

MTH622 Vectors and Classical Mechanics

Question No : [illegible]

The time taken by the particle to complete one oscillation is called

Answer (Please select your correct option)

- amplitude
- time period

Module 103

- frequency
- none of these

MTH622 Vectors and Classical Mechanics

Question No : 27 of 41

Until a force acts on a body, it's velocity is

Answer (Please select your correct option)

- zero
- constant
- increasing
- decreasing

B

MTH622 Vectors and Classical Mechanics

Question No : 29 of 41

Total amount of mass and energy together in a system is

Answer (Please select your correct option)

- increasing
- decreasing
- zero
- constant

D

Question No : [illegible] Marks: 1 (Budg

In a conservative field, the total energy of particle ____________________ throughout the motion.

Answer (Please select your correct option)

- varies rapidly
- does not remain constant
- remains constant
- none of these

Module 115

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MC180

MTH622 Vectors and Classical Mechanics

Question No : 11 of 41 Marks: 1 (Budgeted Time 1 Min)

The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

Answer (Please select your correct option)

- $\frac{bh^3}{12}$
- $\frac{bh^3}{36}$
- $\frac{bh^3}{3}$
- $\frac{bh^3}{4}$

D

http://serverdutes/client/paper.aspx

MTH622 Vectors and Classical Mechanics

Question No : 12 of 41

Moment of inertia of thin rod is given by formula of

Answer (Please select your correct option)

- $\frac{1}{12}(ml^2)$

Module 143, pg82

- mr^3
- $2(mr)$
- $\frac{2}{5}(mr^2)$

MTH622 Vectors and Classical Mechanics

Question No : 18 of 41

Moment of inertia of sphere is given as

Answer (Please select your correct option)

- $2(ml^2)$
- r^2
- $\frac{1}{2}(r)$
- $\frac{2}{5}(mr^2)$

Module 136

Start Time: 10:56 AM

Question No : [illegible] of 41

The moment of inertia of a rectangle base 'b' and depth 'd' about the base will be

Answer (Please select your correct option)

- $\frac{bd^2}{6}$
- $\frac{bd^3}{12}$
- $\frac{db^3}{12}$
- $\frac{bd^3}{3}$

D

Start Time [illegible] AM

16:00

Question No [illegible]   Marks [illegible]

Moment of inertia of any section about an axis passing through its centre of gravity is

Answer (Please select your correct option)

- Maximum
- Minimum
- Depends upon the dimensions of the section
- Depends upon the shape of the section

B

Which of the following is the simplest type of motion along a straight line path. Rectilinear motion (Mod III)

109) An example of path independence is work done by the gravitational force. True (Mod97, Pg38)

110) A curl has Zero divergence (Mod 23, pg 43)

111) If $\nabla \cdot \hat{i}$, then the vector field $\vec{v}$ is non-Solenoidal

112) Unit circle is not a simple closed curve. True

(89) According to Newton's Second law of motion the net force applied on a body is equal to the rate of change of momentum of body. **True** (Mod 117, Pg 36)

Question 7

Consider the force field $F = -kr^3\vec{r}$ Find the potential energy of the given force field

Solution :

Since the field is conservative there exists a potential V such that

$$F = -\nabla V$$

$$F = -kr^3\vec{r} = -k(x^2 + y^2)^{\frac{3}{2}}(xi + yj)$$

$$= -k(x^2 + y^2)^{\frac{3}{2}}xi - k(x^2 + y^2)^{\frac{3}{2}}yj = -\nabla V$$

$$= -\frac{\partial V}{\partial x}i - \frac{\partial V}{\partial y}j - \frac{\partial V}{\partial z}k$$

By comparing, we get

$$\frac{\partial v}{\partial x} = k(x^2 + y^2)^{3/2}x, \quad \frac{\partial v}{\partial y} = k(x^2 + y^2)^{3/2}y \quad \text{and} \quad \frac{\partial v}{\partial z} = 0$$

From which, by omitting the constants, we get

$$V = \frac{1}{5}k(x^2 + y^2)^{5/2} = \frac{1}{5}kr^5$$

<u>Question 10</u>

Linear Momentum

We define the linear momentum ρ of the system as the vector sum of the linear momentum of the individual particles, namely,

$$p = \sum_i p_i = \sum_i m_i v_i$$

From equation (1)

$$r_{cm} = \frac{\sum_{i=1}^{n} m_i r_i}{m} \qquad (1)$$

$$\dot{r}_{cm} = v_{cm} = \frac{\sum_{i=1}^{n} m_i v_i}{m}$$

it follows that

$$p = m v_{cm}$$

that is, the linear momentum of a system of particles is the product of the velocity of the center of mass and the total mass of the system.

Question 11

formulas for translational and rotational K.E. of a rigid body

Solution:

translational formula

$$T_{tr} = \frac{1}{2}Mv^2$$

rotational formula

$$T_{rot} = \frac{1}{2}\omega . L \qquad \text{eq (1)}$$

We known $L = I\omega$

We k putting the value of L in eq (1) we get

$$T_{rot} = \frac{1}{2}\omega . I\omega = \frac{1}{2}I\omega^2$$

Question 12

the moment of inertia of a rectangular plate with sides a and b about an X axis $I_x = \frac{1}{3}Mb^2$ and Y axis $I_y = \frac{1}{3}Ma^2$ Find the moment of inertia along z-axis

Solution:

$$I_z = I_x + I_y$$

$$I_z = \frac{1}{3}Mb^2 + \frac{1}{3}Ma^2$$

$$I_z = \frac{1}{3}M(a^2 + b^2)$$

Questions 13

Calculate the moment of inertia I_{cm} for a uniform rod of length l and mass M rotating about an axis through the center, perpendicular to the rod.

Soution:

In order to calculate the moment of inertia through the center of mass c.m., we use parallel axes theorem.

In a transparent notation

$$I_i = I_{cm} + Md^2 \tag{1}$$

where d is the distance between the origin and the center of mass and $d = {}^{l}/_{2}$.

Also I_i is the moment of inertia of rod about one of its end (which we calculated earlier).

Here

$$I_i = \frac{1}{3}Ml^2 \tag{2}$$

From (1), we have

$$I_{cm} = I_i - Md^2$$

Substituting value from (2) in (1)

$$I_{cm} = \frac{1}{3}Ml^2 - M\left(\frac{l}{2}\right)^2 = \frac{1}{3}Ml^2 - \frac{1}{4}Ml^2$$

$$I_{cm} = \frac{1}{12}Ml^2$$

which is required moment of inertia about its center of mass.

Question No : 34 of 41 **Marks: 2 (Budgeted Time 4 Min)**

Write the expression for total energy of a particle executing Simple Harmonic Motion.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Start Time: 10:50 AM

Question 1

Write the expression for total energy of a particle executing simple harmonic motion.

Ans:

If T is the kinetic energy, V the potential energy and $E = T + V$ the total energy of a simple harmonic oscillator, then we have

$$K.E = T = \frac{1}{2}mv^2$$

and

$$V = \frac{1}{2}kx^2$$

Then the total energy of S.H.M will be

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

MTH622 Vectors and Classical Mechanics

Question No : 33 of 41 MC1

Marks: 2 (Budgeted Time 4 Min)

If moment of inertia of a uniform circular cylinder of length b and radius a about x- axis through the center and perpendicular to the central axis is:

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

nswer (Please click here to Add Answer)

100%

Normal Arial 12

Question 5

If the moment of inertia of a uniform circular cyclinder of length b and radius a about x-axis through the center and perpendicular to the central axis is :

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

Solution:

Due to symmetry we have

$$I_x = I_y = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner, if $I_{xx} = \frac{M}{3}(b^2 + c^2)$

then find I_{yy}, and I_{zz}

Answer (Please click here to Add Answer)

100%

Normal | Arial | 12

Question 4

To find the inertia of uniform solid cuboid (parallelpiped) at one of its corner, $I_{xx} = \frac{M}{3}(b^2 + c^2)$ if Then find I_{yy} and I_{zz}

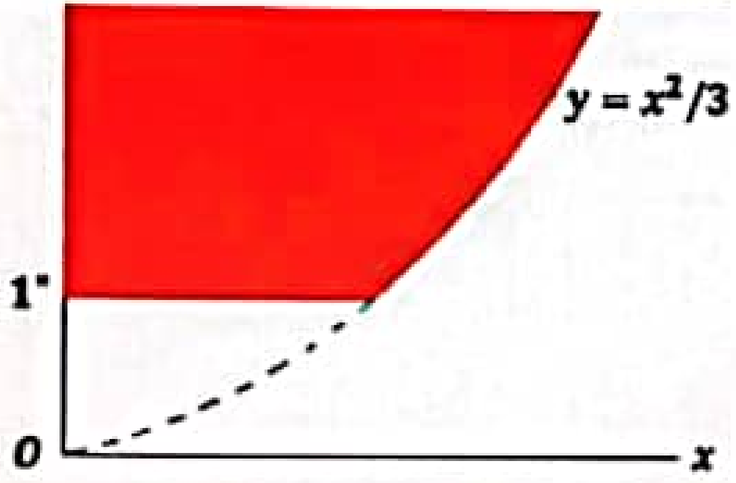
Solution:

Similarly, due to symmetry, we can write

$$I_{yy} = \frac{M}{3}(a^2 + c^2)$$

$$I_{zz} = \frac{M}{3}(a^2 + b^2)$$

Question No : 41 of 41 **Marks: 5**

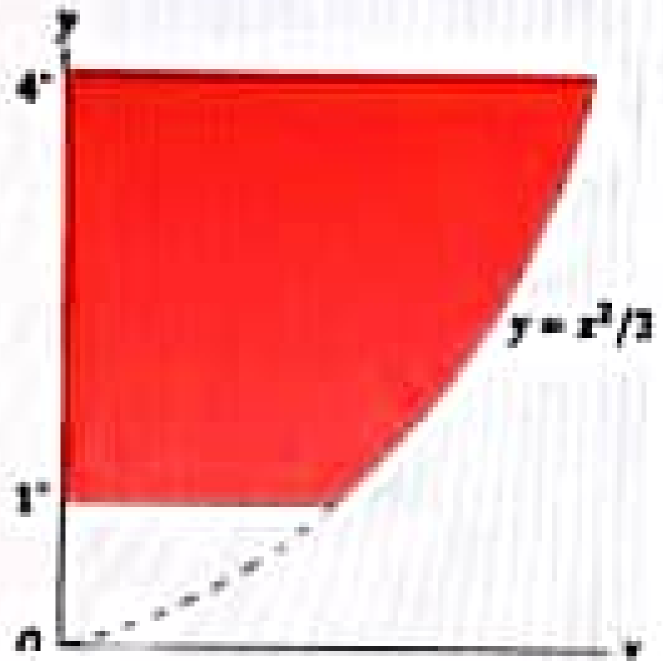


whereas an elementary strip from the shaded area whose Moment of inertia is

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Calculate the moment of inertia of the shaded area given in figure about y-axis.



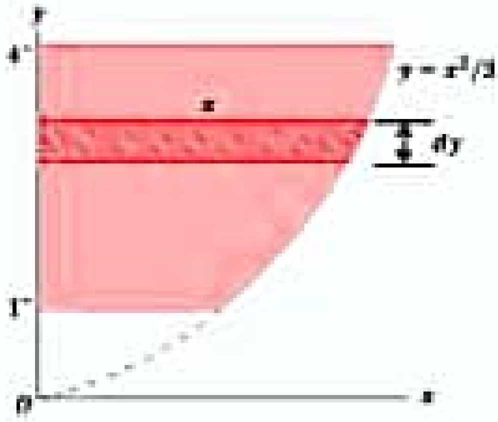
Solution

Here, we have

$$y = x^2/3 \qquad (1)$$

We consider an elementary strip from the shaded area whose M.I is

$$I_{strip} = \frac{1}{3}x^2(x\,dy) = \frac{1}{3}x^3\,dy$$



Then by integration, we obtain the moment of inertia of the whole shaded area

$$I_y = \sqrt{3}\int_1^4 y^{3/2}\,dy = \sqrt{3}\,\frac{2}{5}\left|y^{5/2}\right|_1^4$$

$$= \sqrt{3}\,\frac{2}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}(32 - 1)$$

$$= 21.5\ \text{in}^4$$

Question No : 40 of 41 Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point.

(Let the x-axis be chosen along the length of the rod, with origin at one end point Let *M* and *a* be the mass and length of rod respectively and suppose the rod to be composed of small elements.)

Answer (Please click here to Add Answer)

Normal | Arial | 12 | 100%

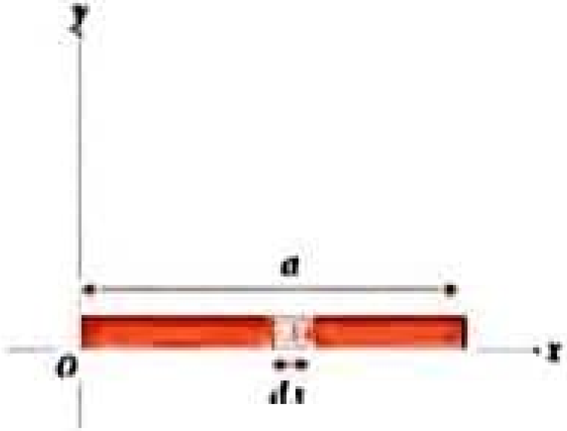
Questions 3

Calculate the moment of inertia of a uniform rod of length l about an axis perpendicular to the rod and passing through an end point

Solution:

Let the x − axis be chosen along the length of the rod, with origin at one end point as shown in the figure. Let M and a be the mass and length of rod respectively. We suppose the rod to be composed of small elements.

Let dm and dx be the mass and the length of the specific element of the rod at a distance x from the end point O.



Then

$$\frac{dm}{dx} = \frac{M}{a}$$

$$\Rightarrow dm = \frac{M}{a}dx$$

Then the moment of inertia of this element about the given axis is

$$I_{element} = dm\,x^2 = \frac{M}{a}x^2dx$$

Hence the moment of inertia of the whole rod will be

$$I = \sum_{\text{all elements}} \frac{M}{a}x^2dx$$

$$= \frac{M}{a}\int_0^a x^2dx$$

$$= \frac{M}{a}\cdot\left|\frac{x^3}{3}\right|_0^a = \frac{M}{a}\frac{a^3}{3}$$

$$= \frac{1}{3}Ma^2$$

MTH622 Vectors and Classical Mechanics

Question No : 39 of 41

Find the moments of inertia of a ring of radius $2a$ about an axis through center.

Answer (Please click here to Add Answer)

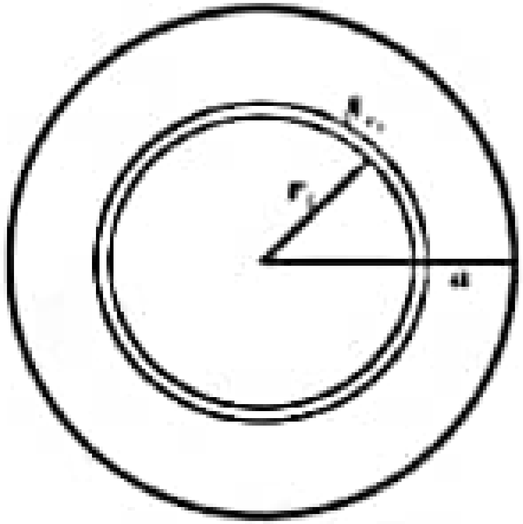
Normal Arial 12 100%

Find the moments of inertia of a ring of radius a about an axis through center

Solution:

Let M be the mass and a the radius of the ring. Then the mass per unit length will be $M/_{2\pi a}$
We regard this ring to be composed of small elements of mass (δm) each of length δs,
we can write it as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi a} \Rightarrow \delta m = \frac{M}{2\pi a}\delta s$$



Moment of inertia of the element about an axis through center O and the perpendicular to the plane of the ring equals $\frac{M}{2\pi a}\delta s\, a^2$.
Therefore the M I of the whole ring will be

$$I_{ring} = \frac{M}{2\pi a}a^2 \sum_{elements} \delta s = \frac{Ma}{2\pi}\int ds = \frac{Ma}{2\pi} \times 2\pi a = Ma^2$$

Hence Ma^2 is the required moment of inertia of the ring

MTH622 Vectors and Classical Mechanics

Question No : 31 of 41

What is a degrees of freedom of the system of rigid bodies.

Answer (Please click here to Add Answer)

Normal | Arial | 12 | 100%

Start Time: 10:56 AM

Question 9

What is a degree of freedom of the system of rigid bodies.

Solution :

The number of coordinates required to specify the position of a system of one or more particles is called the number of degrees of freedom of the system.

For example a particle moving freely in space requires 3 coordinates, e.g. (x, y, z), to specify its position. Thus the number of degrees of freedom is 3.

MTH622 Vectors and Classical Mechanics

Question No : 31 of 41

State Euler's theorem.

Answer (Please click here to Add Answer)

Normal Arial 12

Euler

Theorem Statement

"A rotation of a rigid body about a fixed point of the body is equivalent to a rotation about a line which passes through the point."

The line referred to is called the instantaneous axis of rotation. Rotations can be considered as finite or infinitesimal. Finite rotations cannot be represented by vectors since the commutative law fails. However, infinitesimal rotations can be represented by vectors.

Question No : 32 of 41

Write down the mathematical expression for angular momentum of a single particle

nswer (Please click here to Add Answer)

Normal Arial 12 100%

Angular Momentum

The angular momentum of a single particle is defined as the cross product of linear momentum and position vector of concerned particle.
Mathematically $L = r \times mv$.

Angular Momentum of System of Particles

The angular momentum L of a system of particles is defined accordingly, as the vector sum of the individual angular momentum, namely,

$$L = \sum_{i=1}^{n} r_i \times m_i v_i \tag{1}$$

Question No : [illegible] Marks: 3 (

Write down the expression for moment of inertia of the shaded area shown in figure with respect to x.

nswer (Please click here to Add Answer)

$x = a,\ y = b$, then

$$b = ka^2 \Rightarrow k = \frac{b}{a^2}$$

Substituting the value

of k in (1), we obtain

$$y = \frac{b}{a^2}x^2 \text{ or } x = \frac{a}{\sqrt{b}}\sqrt{y}$$

Now the moment of inertia along x-axis will be

$$I_x = \int_A y^2 dA = \int_0^b y^2(a - x)dy$$

$$= \int_0^b y^2\left(a - \frac{a}{\sqrt{b}}\sqrt{y}\right)dy$$

$$= a\int_0^b y^2 dy - \frac{a}{\sqrt{b}}\int_0^b y^{5/2}dy$$

$$= a\left|\frac{y^3}{3}\right|_0^b - \frac{a}{\sqrt{b}}\left|\frac{2}{7}y^{7/2}\right|_0^b$$

$$= \frac{ab^3}{3} - \frac{a}{\sqrt{b}}\frac{2}{7}b^{7/2}$$

$$= \frac{ab^3}{3} - \frac{2ab^3}{7}$$

$$I_x = \frac{ab^3}{21}$$

Question No : 33 of 41

If $e_\rho = \cos\varphi\, \hat{i} + \sin\varphi\, \hat{j}$ and $e_\varphi = -\sin\varphi\, \hat{i} + \cos\varphi\, \hat{j}$ then prove that

$$\frac{de_\rho}{dt} = \dot{\varphi} e_\varphi \quad \text{and} \quad \frac{de_\varphi}{dt} = -\dot{\varphi} e_\rho$$

where dots denote differentiation with respect to time t.

Answer (Please click here to Add Answer)

100%

Normal | Arial | 12

Problem Statement

Prove

$$\frac{de_\rho}{dt} = \varphi \dot{e}_\varphi$$

$$\frac{de_\varphi}{dt} = -\dot{\varphi} e_\rho$$

where dots denote differentiation with respect to time t.

Solution

We have

$$e_\rho = \cos\varphi\, \imath + \sin\varphi\, \jmath$$

and

$$e_\varphi = -\sin\varphi\, \imath + \cos\varphi\, \jmath$$

Then

$$\frac{de_\rho}{dt} = \frac{d}{dt}(\cos\varphi\, \imath + \sin\varphi\, \jmath)$$

$$= -\sin\varphi\, \dot{\varphi}\imath + \cos\varphi\, \dot{\varphi}\jmath = \dot{\varphi}(-\sin\varphi\, \imath + \cos\varphi\, \jmath) = \dot{\varphi} e_\varphi$$

$$\frac{de_\varphi}{dt} = \frac{d}{dt}(-\sin\varphi\, \imath + \cos\varphi\, \jmath)$$

$$= -\cos\varphi\, \dot{\varphi}\imath - \sin\varphi\, \dot{\varphi}\jmath = -\dot{\varphi}(\cos\varphi\, \dot{\varphi}\imath + \sin\varphi\, \dot{\varphi}\jmath) = -\dot{\varphi} e_\rho$$

MTH622 ... tors and Classical Mechanics

Marks: 2 (Budgeted Time Min)

Question No :

If the surface area of the ring is $2\pi r\,dr$ and surface area of annulus is $\pi(R_2^2 - R_1^2)$, then find the mass dm of ring. Whereas mass of disc is M.

Answer (Please click here to Add Answer)

100%

The Surface area of the ring is

$$\text{Area} = (2\pi r)dr = 2\pi r dr$$

Since the surface area of the

annulus is

$$\pi(R_2^2 - R_1^2)$$

Therefore, we can have

$$\frac{dm}{M} = \frac{2\pi r dr}{\pi(R_2^2 - R_1^2)}$$

$$dm = \frac{2rdr}{(R_2^2 - R_1^2)}M$$

Question No : 35 of 47

Write Necessary and sufficient conditions for a Conservative Force Field

Answer (Please click here to Add Answer)

100%

Normal Arial 12 B I U

Conservation of Energy for a System of Particles

Statement

"The law of conservation of energy describes that the net energy of an isolated system remains conserved. Energy can neither be created nor destroyed; rather, it transforms from one form to another."

Theorem Statement

(Principle of conservation of energy)

In case of conservative force field, the total energy is a constant. i.e.,

If T is for kinetic energy and V is for potential energy, then the total energy E is

$E = T + V = \text{constant}$

MTH622 Vectors and Classical Mechanics

Question No : 36 of 41 Marks: 3 (Bud

If $r = a\cos\omega t\, i + b\sin\omega t\, j$ and $p = -am\omega\sin\omega t\, i + bm\omega\cos\omega t\, j$ then find the angular momentum.

Question 8

if $r = a\cos\omega t\, i + b\sin\omega t\, j$. and $p = -am\omega\sin\omega t\, i + bm\omega\cos\omega t\, j$

then find angular momentum

Solution :

We known that

$$r \times p$$

$$= \begin{vmatrix} i & j & \hat{k} \\ a\cos\omega t & b\sin\omega t & 0 \\ -am\omega\sin\omega t & bm\omega\cos\omega t & 0 \end{vmatrix}$$

$$= [abm\omega\cos^2\omega t + abm\omega\sin^2\omega t]\hat{k}$$

$$= abm\omega\hat{k}$$

$$\tau = \frac{d\Omega}{dt} = \frac{d}{dt}(r \times p) = 0$$

MTH622 Vectors and Classical Mechanics

Question No : [illegible] **Marks: 5 (Budge**

Find the moment of inertia of a solid circular cylinder of radius a, mass M and the height of the cylinder h about t
cylinder. Whereas moment of intertial of one shell be

$$I_{shell} = r^2 \frac{2M}{a^2} r\,dr = \frac{2M}{a^2} r^3 dr$$

(the mass of one shell is dm, height is same as h, thickness be dr and the radius be r and)

$$I_{shell} = r^2 \frac{2M}{a^2} r\,dr = \frac{2M}{a^2} r^3 dr$$

Thus the total moment of inertia of the solid circular cylinder will be

$$I_{cylinder} = \int_{r=0}^{a} I_{shell}$$

$$I_{cylinder} = \int_{r=0}^{a} \frac{2M}{a^2} r^3 dr$$

$$= \frac{2M}{a^2} \int_{r=0}^{a} r^3 dr$$

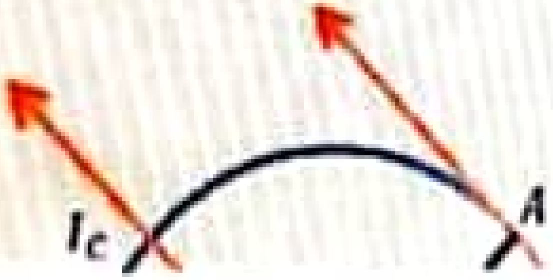
$$= \frac{2M}{a^2} \left| \frac{r^4}{4} \right|_0^a = \frac{2M}{a^2} \frac{a^4}{4}$$

$$I_{cylinder} = \frac{1}{2} M a^2$$

is the M.I of solid cylinder.

Question No : [illegible] Marks: 5 (Budgeted Time Min)

Use the parallel axis theorem to find the moment of inertia of a solid circular cylinder about a line on the surface of the cylinder and parallel to axis of cylinder.



Answer (Please click here to Add Answer)

By parallel axis theorem

$$I_A = I_c + Ma^2 \qquad (1)$$

Since I_C which is the moment of inertia of a solid circular cylinder about an axis passing from the center of mass is defined by

$$I_C = \frac{1}{2} Ma^2 \qquad (2)$$

where a is the radius of a solid circular cylinder.

By substituting equation (2) in equation (1) we have

$$I_A = \frac{1}{2} Ma^2 + Ma^2$$

$\Longrightarrow$

$$I_A = (\frac{1}{2} + 1) Ma^2$$

$$I_A = \frac{3}{2} Ma^2$$

MTH622 Vectors and Classical Mechanics

Question No : 37 of 48 **Marks: 3 (Budgeted Time Min**

Write down the expression for moment of inertia of a uniform hemisphere of radius a about its axis of symmetry, where

$$dV = dr(rd\theta)(r\sin\theta d\varphi) = r^2\sin\theta\, dr d\theta d\varphi$$

Note: Only write the expression. Do not evaluate the integrals.

Answer (Please click here to Add Answer)

100%

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$$I_{zz} = \frac{2}{5} M a^2$$

which is required expression for moment of inertia of hemisphere.

Question No : 36 of 41 **Marks: 3 (Budgeted Time 5 Min)**

Consider a small element of a uniform elliptic plate has moment of interia $I = \delta m y^2$ Write down the expression for moment of interia of whole plate.
(Note: Do not solve the expression)

Answer (Please click here to Add Answer)

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36

where for elliptical plate, we have

$$I_x = \frac{1}{4}ab^3\pi \times \frac{M}{\pi ab} = \frac{1}{4}Mb^2$$

$$\rho = \frac{M}{\pi ab}$$

is the required expression for M.I.

Question No : 38 of 41 Marks: 3 (Budgeted Time 6 Min)

What is the impulse developed by a mass of 100 gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec ?

Answer (Please click here to Add Answer)

100%

12

Q. What is the impulse developed by a mass of 100 gm which changes its velocity from $\hat{i}+\hat{j}+\hat{k}$ to $2\hat{i}+2\hat{j}+3\hat{k}$ m/sec?

Sol:- $m = 200$, $v_1 = \hat{i}+\hat{j}+\hat{k}$, $v_2 = 2\hat{i}+2\hat{j}+3\hat{k}$

we know that

$$\text{impulse} = P_2 - P_1$$

$$= mv_2 - mv_1$$

$$\text{impulse} = m(v_2 - v_1)$$

$$= 200\{(2\hat{i}+2\hat{j}+3\hat{k}) - (\hat{i}+\hat{j}+\hat{k})\}$$

$$= 100\{(2\hat{i}-\hat{i}) + (2\hat{j}-\hat{j}) + (3\hat{k}-\hat{k})\}$$

$$= 100(\hat{i}+\hat{j}+2\hat{k})$$ Ans

Question No : 37 of 41 Marks: 3 (Budgeted Time 6 Min)

If a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}$ relative to the fixed coordinate system, and $\vec{A} = \sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k}$ then find dA/dt with respect to the fixed point system.

Where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t\,\hat{\imath} + \cos t\,\hat{\jmath} - 2e^{-2t}\hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t)\hat{\imath} - (-2e^{-t} + 3\sin t)\hat{\jmath} + (\cos t + 2\sin t)\hat{k}$

Answer (Please click here to Add Answer)

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Q. If a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ relative to the fixed coordinates system, and $A = \sin t\,\hat{i} + \cos t\,\hat{j} + e^{-t}\hat{k}$ then find $\frac{d\vec{A}}{dt}$ with respect to the fixed point system where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t\,\hat{i} + \cos t\,\hat{j} - 2e^{-2t}\hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t)\hat{i} - (-2e^{-t} + 3\sin t)\hat{j} + (\cos t + 2\sin t)\hat{k}$

Sol:-

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \vec{\omega} \times \vec{A}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = -\sin t\,\hat{i} + \cos t\,\hat{j} - 2e^{-2t}\hat{k} + (4e^{-2t} - 3\cos t)\hat{i} - (-2e^{-t} + 3\sin t)\hat{j} + (\cos t + 2\sin t)\hat{k}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = (-\sin t + 4e^{-2t} - 3\cos t)\hat{i} + (\cos t + 2e^{-t} - 3\sin t)\hat{j} + (-2e^{-2t} + \cos t + 2\sin t)\hat{k}$$ Ans

Question No :35 of 41

Marks: 3 (Budgeted Time 6 Min)

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

Answer (Please click here to Add Answer)

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Important notes Mth-622

Final term for MCQ'S

Made by:- "**VU Life Study**"

1. **Path Independence**
- Independence of path is defined as that the **work done** by an object remains **same** between **initial point** and the **final destination** of the particle, it does not matter which path is taken by the object.
- "A force field is said to **conservative** if the total work done by the particle moving along a curve is **independent** of the path taken by the particle and depend upon the end points of the **curve** only."

1. **Necessary and sufficient conditions for a Conservative Force Field:-**
- A force field F is conservative if and only if there exists a **continuously differentiable** scalar field V such that $\vec{F}=-\nabla V$ or, equivalently, **if and only if curl** $\vec{F}=\nabla\times\vec{F}=0$ **identically**.
- A continuously differentiable force field F is **conservative** if and only if for **any closed nonintersecting curve C** (simple closed curve) $W=\oint_c \vec{F}.d\vec{r}=0$
- The total work done in moving a particle around **any closed path** is **zero.**

2. **Examples of Conservative Forces**
- **Gravitational force** , **Elastic spring force** is example of conservative force.
- The work done of a particle moving along a **closed path is zero** and the force which causes such **motion is conservative**.
- Forces that cannot be expressed in the term of a **potential energy** function are called **non-conservative** forces.
- The forces that **do not store** energy are called **nonconservative** or **dissipative** forces.
- If there is **no scalar** function V such that $F=-\Delta V$ [or, **equivalently, if** $\boldsymbol{V\times F=0}$], then F is called **a non-conservative** force field.
- **Friction,** the impulse (**time dependent** force) is also a non-conservative force.
- **Simple Harmonic Motion (SHM)** is a particular type of **oscillation** and periodic motion in which restoring force of an object is **directly proportional** to the displacement of the object acting in opposite direction of displacement.
- Mathematically, the **restoring force** $\boldsymbol{F}$ is given by $F_R=-\boldsymbol{kx}$.

  where the **subscript** $\boldsymbol{R}$ represents the restoring force and $\boldsymbol{k}$ is the constant of **proportionality** often called the **spring constant** or **modulus of elasticity.**
- In Newtonian mechanics, by Newton's second law we have eq. of S.H.M
- $F=ma=m\dfrac{d^2x}{dt^2}=-kx$ or $\boldsymbol{m\ddot{x}+kx=0}$.

- This vibrating system is called a simple harmonic oscillator or linear harmonic oscillator.
- **Mass on a spring:-**
  An object of mass m linked to a spring of spring constant k represents the simple harmonic motion **in closed region.**
- The equation representing the period for the mass attached on a spring
- $\boldsymbol{T} = 2\pi\sqrt{\frac{m}{k}}$
- The above equation expresses that the **time period** of oscillation is **independen**t of **amplitude** as well as the acceleration.
- **Simple pendulum:-**
  The movement of the mass attached to a simple pendulum is considered as simple harmonic motion.
- The time period of a mass m attached to a pendulum of length l with gravitational acceleration g is given by T= $2\pi\sqrt{\frac{l}{g}}$.
- **Amplitude**:-
- Amplitude is defined as **the maximum distance** covered by the oscillating body **in one oscillation or length** of a wave measured from its **mean position**.
- The amplitude of a pendulum is **one-half the distance** that the mass covered in moving from **one terminal to the other**.
- The vibrating sources generate **waves,** whose amplitude is **proportional** to the amplitude of the **vibrating source**.
- Time period is **minimum time** required by a oscillation system to complete **its one cycle** of oscillation of the specific system.
- It is denoted by **T** and measured in **seconds**.
- The **frequency (f)** of an oscillatory system is the number of oscillations pass through a **specific point** in **one second**.
- It is measure in **hertz (Hz).**
- The frequency of **S.H.M** can be calculate by using the following relation $f = \frac{1}{T}$.
- If **T is the kinetic energy**, **V the potential energy** and $\boldsymbol{E} = \boldsymbol{T} + \boldsymbol{V}$ the total energy of a simple harmonic oscillator, then we have $K.E = T = \frac{1}{2}mv^2$

  and $V = \frac{1}{2}kx^2$.

  Then the **total energy of S.H.M** will be $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$.
- The forces acting on a harmonic oscillator are called **damping forces** which tend to **decrease the amplitude of the successive oscillations** or simply force apposing the motion.
- The damping force is **proportional** to the **velocity.**

- Mathematically, $Fd = -bv = -b\dot{x}$
- where d represents the **damping force** and b is the **damping coefficient**.
- The **negative** sign shows that that direction of Fd is **opposite** to the **velocity v.**
- $\frac{b}{m} = 2\zeta,$
- $\sqrt{\frac{k}{m}} = \omega 0$
- The equation of damped harmonic oscillator can be written as $\ddot{\ }+2\zeta\dot{x}+\omega_0^2\,\mathbf{x} = \mathbf{0}$.
- The theorem, called **Euler's theorem**, is fundamental in the motion of **rigid bodies**.
- "A rotation of a **rigid body** about a **fixed point** of the body is **equivalent** to a rotation about a line **which passes through the point**."
- The line referred to is called **the instantaneous axis of rotation**.
- **Rotations** can be considered as **finite or infinitesimal**.
- **Finite** rotations **cannot** be represented by **vectors** since the **commutative law fails.**
- **Infinitesimal** rotations can be represented by **vectors.**
- **Chasle's theorem** states that the most general **rigid body displacement** can be produced by a **translation along a line** (called its screw axis) followed (or preceded) by a rotation about that line.
- A **rigid** body has **six degrees** of **freedom**.
- By **Euler's theorem**, **three** of these are associated with **pure rotation.**
- The remaining **three** must be associated with **translation**.
- To describe the general motion of a rigid body, think of the general motion as translation of a fixed point O in the body to a point O' followed by the rotation about an axis through O'.
- **Kinematics** is the branch of mechanics deals with the **moving objects** without reference to the forces which cause the **motion**.
- Some **features** of **rigid body motion** are:-
- Displacement.
- Position.
- Velocity.
- Linear Velocity & Angular Velocity.
- Linear Acceleration & Angular Acceleration.
- Motion of a Rigid Body (Translation & Rotation).
- In **Space** this is closely related to the concepts of point, **position**, ' **direction** and **displacement**'. Measurement in space involves the concepts of **length or distance**, with which we assume familiarity.
- **Units** of length are **feet**, **meters**, **miles**, etc.
- In **Time** this concept is derived from our experience of having **one event** taking place **after, before** or **simultaneous with another** event.
- **Units** of time are **seconds, hours, years**, etc.

- In **Matter** Physical objects are composed of **"small bits of matter"** such as **atoms** and **molecules**
- A measure of the **"quantity of matter"** associated with a particle is called its **mass**.
- **Units** of **mass** are **grams, kilograms,** etc.
- Unless otherwise stated we shall assume that the mass of a particle **does not change** with **time.**
- When a **moving particle** remains on a **single straight line**, the motion is said to be **rectilinear.**
- The **general** equation of **motion** is then $= ma \Longrightarrow (x,\dot{x}) = m\ddot{x}$.
- Rectilinear motion of a body is defined by considering the **two point** of a body covered the **same distance** in the **parallel** direction.
- An **exampl**e of linear motion is an **athlete running** g along a **straight** track.
- The rectilinear motion can be of two types:-
  i. Uniform rectilinear motion.
  ii. Non uniform rectilinear motion.
- **Uniform rectilinear** motion is a type of motion in which the body moves with **uniform velocity** or **zero acceleration.**
- **Non uniform** rectilinear motion is such type of motion with **variable velocity** or **nonzero acceleration.**
- **Uniformly accelerated** rectilinear motion is a special case of **non-uniform** rectilinear motion along a line is that which arises when an object is subjected to **constant** acceleration. This kind of motion is called **uniformly accelerated motion**.
- Uniformly accelerated motion is a type of motion in which the velocity of an object changes by an **equal amount in every equal intervals of time**.
- An example of uniformly accelerated body is **freely falling** object in which the amount of gravitational acceleration **remains same**.
- $F = mg$
- The motion of a particle moving in a **curved path** is called **curvilinear motion**.
- **Example**: A stone thrown into the air at **an angle**.
- **Curvilinear motion** describes the motion of a moving particle that confirms to **a known or fixed curve.** The study of such motion involves the use of **two co-ordinate** systems, the first being **planar motion** and the latter being **cylindrical motion.**
- **Tangential** and **normal** unit vectors are usually denoted by $\vec{e}_t$ *and* $\vec{e}_n$ respectively.
- **Velocity of Curvilinear motion:-**
- If the tangential and normal unit vectors are $\vec{e}_t$ *and* $\vec{e}_n$ respectively, then the velocity will be $\vec{v} = \frac{ds}{dt}\vec{e}_t$

  You have already learnt that $\boldsymbol{v} = \mathbf{vT}$.
- **Acceleration of Curvilinear motion**:-
- If the tangential and normal unit vectors are $\vec{e}_t$ *and* $\vec{e}_n$ respectively, then the acceleration will be

- $a^{\to} = \frac{d^2 s}{dt^2}\vec{e}_t + \frac{(ds/dt)^2}{\rho}\vec{e}_n$
- You have already learnt that
- $a = T\frac{dv}{dt} + \frac{v^2}{r}N$
- Examples of Curvilinear motion:-
- A stone thrown into the air at **an angle.**
- A car driving along a **curved road**.
- **Throwing paper** airplanes or paper **darts** is an example of curvilinear motion.
- **e1,2 and e3** are **mutually perpendicular** and so the cylindrical coordinate system is **orthogonal.**
- **Introduction to Projectile**
- If a ball is thrown from **one person to another** or an object is dropped from a moving plane, then their path of traveling/motion is often called a **projectile.**
- If **air resistance** is **negligible**, a projectile can be considered as a **freely falling body** so the Eq of motion will be $m\frac{d^2r}{dt^2} = -mg$ or $\frac{d^2r}{dt^2} = -g$.
- "The law of conservation of energy describes that the **net energy** of an isolated system **remains conserved**. Energy **can neither be created nor destroyed**; rather, it transforms from one form to another."
- In case of conservative force field, the **total energy** is a **constant**.
- If T is for kinetic energy and V is for potential energy, then the total energy E is $E = T + V$ = constant .
- Impulse is a special type of force defined by applying the integral of a force F, over the time interval, t, for which it acts on the body.
- Impulse is a **directional (vector)** quantity in the **same direction** of force as force is also a **directional quantity**.
- When Impulse is applied to a rigid body, it results a **corresponding vector** change in its **linear momentum** along the **same direction**.
- The SI unit of impulse is the **newton second** ($N \cdot s$), and the dimensionally equivalent unit of momentum is the **kilogram meter per second** ($kgms{-1}$).
- Torque is defined as the turning effect of a body. It is trend of an acting force due to which the rotational motion of a **body changes**. It is also called **twist** and **rotational force** on an object.
- Torque is defined as the cross product of the force vector to the distance vector, which causes rotational motion of the body. $\boldsymbol{\tau} = \vec{r} \times \vec{F}$
- The magnitude of **torque** depends upon the **applied force.**
- The length of the lever arm connecting the axis to the point where the force applied, and the angle between the force vector and the length of lever arm. Symbolically we can write it as: $\boldsymbol{\tau} = |\boldsymbol{r}||\boldsymbol{F}|\sin\boldsymbol{\theta}$

- Torque is a **vector quantity** implies that it has direction as well as **magnitude**.
- The SI unit for torque is the **newton meter (Nm).**
- The **direction** of torque can be approximate using **Right Hand Rule.**

- ## Rigid bodies:-
- When a force is applied to an object/ system of particles, and if the object maintains its **overall shape**, then the object is called a **rigid body.**
- Gap between **two fixed points** on the rigid body remains **same regardless** of external forces exerted on it.
- We can neglect the **deformation** of such bodies.
- A rigid body usually has **continuous** distribution of mass.
- ## Definition of Elastic Bodies:-
- When a force is applied to a system of particles, **it changes the distance** between individual particles. Such systems are often called **deformable** or elastic bodies.
- Examples
- A **spring** and **rubber band** are some common examples of **elastic bodies.**
- A **wheel** is a common example of **rigid body**.
- **Properties of Rigid Bodies**
- The number of coordinates required to specify the position of a system of **one or more particles** is called the number of **degrees of freedom** of the system. For example a particle moving freely in space requires 3 coordinates, e.g. (x, y, z), to specify its position. Thus the number of **degrees of freedom is 3.**
- A displacement of a **rigid body** is a direct change of position of its particles.
- Translational motion is the displacement of all particles of the body by the same amount and the line segment joining **the initial and the final position** of the particles represented by parallel vectors. Examples of translational motion are particles freely falling down to earth and the motion of a bullet fired from a gun.
- Circular motion of a body about a **fixed point** or axis is called **rotation.**
- If during a displacement the points of the rigid body on some line **remains fixed** and all other are **displaced** through the **same angle**, then this displacement is called **rotation**.
- A rigid performs rotations around **an imaginary line** called a rotation axis.
- If the axis of rotation passes through the **center of mass** of the rigid body then body is said to be **spin** or **rotate upon itself.**
- If a body rotates about **some external fixed point** is called **revolution** or **orbital motion** of the rigid body.
- The example of revolution is the rotation of **earth around sun and motion of moon around sun.**
- Rotational motion **concerns** only with **rigid bodies**. The reverse rotation of a body (inverse rotation) is also a rotation.
- A **wheel** is common examples of rotation.

- Translational motion along the given **fixed plane** and **rotational motion** about a **suitable** axis **perpendicular** to the plane.
- This fixed axis is specifically chosen to pass through the **center of mass** of the rigid body.
- The axis about which the rigid body rotates is called **instantaneous axis of rotation**, where this axis is **perpendicular** to the plane.
- The point **where instantaneous axis meets the fixed plane** along which the body performs translation motion is described as the instantaneous **center of rotation.**
- The center of mass (c.m.) or centroid of system of particles is a **hypothetical particle** such that if the entire mass of the system were concentrated there, the **mechanical** properties would **remain the same.**
- In particular expression of linear momentum, angular momentum and kinetic energy assume **simpler or more convenient forms** when referred to the coordinated of this hypothetical particle and the equation of motion can be reduced to simpler equation of a **single particle.**
- If a system **experiences no external force**, the center-of-mass of the system will remain **at rest**, or will move at **constant velocity** if it is already moving.
- If there is an **external force**, the center of mass accelerates according to $\boldsymbol{F = ma}$.
- Basically, the center-of-mass of a system can be treated as a **point mass**, following Newton's Laws.
- If an object is **thrown into the air**, different parts of the object can follow **quite complicated paths**, but the center-of-mass will follow a **parabola.**
- If an object explodes, the different pieces of the object will follow seemingly **independent paths** after the explosion. The center of mass, however, will keep doing what it was doing before the explosion. This is because an explosion involves only **internal forces**.
- The moment of inertia of a rigid body is a property which depends upon its **mass and shape**, (i.e. the mass distribution of the body) and determines its behavior in rotational motion.
- In rotational motion, the moment of inertia plays the **same** role as the mass in linear motion.
- Formally the moment of inertia I of the particle of mass m about a line is defined by $I = md^2$ where d is the perpendicular distance between the particle and the line (called the axis).
- Radius of gyration specifies the distribution of the elements of body around the axis in terms of the mass moment of inertia.
- When a rigid body is rotating about a fixed point O, the **angular velocity** vector ω and the **angular momentum** vector L (about O) **are not in general in the same** direction.
- Each point in the body there exists distinct directions, which are fixed relative to the body, along which the **two vectors** are **aligned** i.e. **coincident.** Such directions are called **principal directions** and the axes along them are referred to as **principal axes of inertia.**
- The corresponding moments of inertia are called principal moments of inertia.

- If the principal axes at each point of the body exist, then their **orthogonality** can be proved by stating that axes relative to which product of inertia are **zero** are the principal axes.
- Dynamics is the branch of mechanics deals **with forces and relationship** fundamentally to the motion but sometimes also to the **equilibrium of bodies.**
- These will be two types of forces, **internal and external**. Internal forces act between particles of the system; all other are **external**.
- The rigid bodies are a system of particles in which the position of particles is **relatively fixed.**
- An actual rigid body consists of **very large number** of particles and therefore we may suppose that there is a **continuous distribution** of mass.
- If (r) denotes density of a volume element dV, surrounding the point r, then the mass of this element will be $(r)V$.
- The **angular momentum** of a single particle is defined as the **cross product** of linear momentum and position vector of concerned particle. Mathematically $= r \times mv$.
- The time rate change of angular momentum in the absence of some external forces is **zero**. Mathematically, we can write $\boldsymbol{dL/dt = 0 \Rightarrow L = \text{constant}}$.
- The time **rate of change of the angular** momentum of a system is equal to the total moment of all the **external forces** acting on the system.
- If a system is isolated, then $\boldsymbol{N = 0}$, and the **angular momentum** remains **constant** in both **magnitude** and **direction**:
- Kinetic energy is the energy produced in any body during **its motion**. It is equal to the **half** of the product of **mass and square** of the velocity of the moving body.
- **Solid Circular Cylinder** :-
- We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder $I = \frac{1}{2}Ma^2$.
- **Hollow Circular Cylinder:-**
- We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder. We consider the thickness of Wall of cylinder is negligible. $I = Ma^2$
- **Solid Sphere:-**
- We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter. $I = \frac{2}{5}Ma^2$.
- **Hollow Sphere:-**
- We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter. We consider the thickness of sphere is negligible. $I = Ma^2$
- **Rectangular Plate:-**
- We consider sides of length $\boldsymbol{a}$ and $\boldsymbol{b}$, and mass M about an axis perpendicular to the plate through the center of mass. $I = \frac{1}{12}M(a^2 + b^2)$
- **Thin Rod:-**

- We assume the length of rod is ***a*** and mass M about an axis perpendicular to the rod through the center of mass. $I = \frac{1}{12}Ma^2$
- **Triangular Lamina:-**
- We assume the height ***h*** and mass M of lamina. $I = \frac{1}{6}Mh^2$.
- **Right Circular Cone:-**
- We assume the radius of the circular cone is a and mass M. $I = \frac{3}{10}Ma^2$.
- The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the sum of the moment of inertia about two mutually perpendicular axis lying in the plane of the body and meeting at the common point with the given axis.
- The axes relative to which **product** of inertia are **zero** are called the principal axes and the moment of inertia along these axes are called **principal moment of inertia**.
- For a rigid body, there exist a set of **three mutually orthogonal** axes called principal axes relative to which the product of inertia are **zero** $\vec{\omega}$ and L are considered along the **same** direction.
- This system will have a **non-trivial** solution, i.e. ($\vec{\omega} \neq 0$), if the determinant of the matrix of coefficients is **zero.**
- A 3×3 symmetrical matrix has **three** real eigenvalues, which may be **distinct** or **repeated.**
- The eigenvectors of a **symmetrical** matrix corresponding to distinct eigenvalues are **orthogonal.**
- It is always possible to find three mutually **orthogonal** eigenvectors for a 3×3 symmetric matrix, whether the eigenvalues are **distinct or repeated.**
- The principal of moment of inertia are **always real** numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where mi and di are both **real.**
- When all **three** principal moments I_1, I_2, I_3 are **distinct**, then by the theorem 2, three mutually **orthogonal principal** axes can be found.
- When $I_1 = I_2$ but $I_1 \neq I_3$, then by the theorem 3, we can still determine **three mutually orthogonal** principal axes, because the inertia matrix is **symmetric**.
- In many instances a body possesses **sufficient symmetry** so that at least **one principal** axis can be found by inspection, i.e;the axis can be chosen so as to make **two of the three** products of inertia **vanish.**
- We consider a plane **rigid body** i.e. a 2D body; for example a plate of **uniform thickness**. Such a system can be regarded as a **coplanar distribution** of mass.
- Since there are **three mutually orthogonal** principal axes, one of them must be **perpendicular** to the plane of the body.
- The other **two axes** will lie in the plane of **lamina**.

- Suppose a rigid body has **no axis** of symmetry. Even so, the **tensor** that represents the moment of inertia of such a body, is characterized by a **real, symmetric 3×3** matrix that can be **diagonalized.** The resulting diagonal elements are the values of the principal moments of inertia of the rigid body.

- The axes of the coordinate system, in which this matrix is diagonal, are the principal axes of the body, because all products of inertia have **vanished.**
- Thus, finding the principal axes and corresponding moments of inertia of any rigid body, **symmetric or not**, is virtually the same as to **diagonalzing** its moment of inertia matrix.
- The solutions are **not independent**. They are subject to the constraint $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
- In other words the resultant vector e_1 specified by these components is a **unit vector**.

Thank You..!
Remember me in ur prayers.

MTH 622 (09-09-2018)

1. If surface area of ring is $2\pi r\,dr$ and surface area of of annulus $\pi(R_2^2 - R_1^2)$. Then find mass dm of ring where mass of disc is M

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2. State rotatory axis Theorem
3. Write necessary conditions for a conservative force field.
4. Write formula for transitional and rotational kinetic energy of a rigid body
5. Calculate M.I of small portion of hoop of mass 'δm' about an axis through centre perpendicular to plane of a ring. Where 'M' is mass of Hoop and 'r' is radius. Where density of hoop is $\rho = \dfrac{M}{2\pi r\,dr}$
6. $I_x = \frac{1}{3} M b^2$ along x-axis

   $I_y = \frac{1}{3} M a^2$ along y-axis

   Find Moment of Inertia about z-axis
7. $F = xz^2\hat{i} - 3yz\hat{j} + y^2z\hat{k}$ is Non-Conservative proove
8. ایک سوال کی statement تو نہیں / یاد ہے اس کا sketch بنانا تھا

   $m_1 = 7kg$ $v_1 = 2m/sec$
   $m_2 = 4kg$
   $u_1 = 4m/sec$ v_2 Find krna tha
   $u_2 = 4m/sec$
9. State Chaseles Theorem

Questions (41)
Q = objective (30)
Q = 2 marks (4)
Q = 3 marks (4)
Q = 5 Marks (3)

Total Mark (65)

objective: Irrotational Flow
(i) $\nabla \times \vec{V} = 0$ (ii) $\nabla \times \vec{V} \neq 0$
(iii) $\nabla \cdot \vec{V} = 0$ (iv) None

objective: A vector is called irrotational if
(i) $\nabla V = 0$ (ii) $\nabla \cdot V = 0$ (iii) $\nabla \times V = 0$ (iv)

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Date 26-06-18

MTH 622 Mid Term paper

① $\vec{A} = xz^2\hat{i}$ Find $\nabla \cdot \vec{A}$

② $\vec{r} = re_r$ Find velocity of particle.

③ $\phi = 3xyz$ Find $\nabla^2\phi$

④ Find unit normal vector $\hat{T}$

$\vec{r} = 2t\hat{i} + 4t\hat{j} + \hat{k}$

⑤ Find directional derivative

$\phi = 3x^2z$ at $(2,1,1)$

in the direction of $4\hat{i} - 3\hat{k}$

⑥ Find work done

$m = 100kg$, $v_1 = 1200 m/s$

$v_2 = 1900 m/s$

Mth622 my paper
Long questions
1)line integral 5 marks
2) newton 2nd law of motion
5 marks

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short
3)newton 2nd law of motion
definition 2mrks
4) line integral values are
given 2 mrks
5)evluate A.B then find
integral 3 mrks
6) check line is depend or
independent values and
limits are given 3 mrks

12:21 pm

Thank you

Sadia

1)r is given...find v

and acceleration.

2)a man in a lift ma

64kg..having acce

0.35ms-2.Find for

man exerts on ear

3)integral 0 to 2 li

Mcqs were easy...
Find work done.2 marks
Find curl 2
Find divergent at given points 3
integral dependent or independent prove integrl given tha.3
Prove.
del. (A x r)=r. (Del x A) 5 marks
Find work done.. mss nd initial final velocity given 5 marks

Math 622 25/6/18

* State Green's Theorem

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* Normalize $3x+2y+z=4$

* If $A = xz\hat{i}^2 - 2xy\hat{j} + yz^2\hat{k}$ then find curl of A

* $\vec{r}$, mass and momentum given that Force field find karni thi

* A ki value given thi and divergence theorem Prove karna tha.

Mth622
Mcqs easy thy tough ni thy
1. Divergence theorem ka question
2. Stocks theorem ka question tha
3. Irrotational proof tha 1 vector ko
4. Newton 2nd law of motion
Baqi 2 question me proof thy 2 choty choty

Mth622
Objectives conceptual
1.newton second law of motion
2.del. A=O AND del*A=0 ... Justify
2.A was given.. Find del. A
3.FOrce find mas and acceleration givn thi.
4.line integral find
5.Stoks law

Mth622
Surface integrl k 2 que
Ik directional derivative

Aa

ko
4. Newton 2nd law of motion
Baqi 2 question me proof thy 2 choty choty

Mth622
Objectives conceptual
1.newton second law of motion
2.del. A=O AND del*A=0 ... Justify
2.A was given.. Find del. A
3.FOrce find mas and acceleration givn thi.
4.line integral find
5.Stoks law

Mth622
Surface integrl k 2 que
Ik directional derivative
Ik koi normal surface wala tha
Volume integralk long que tha
Ik green theroem sa que tha
Write statemt of green theroem
Mcqs conceptual th

1)- Define Green's Theorem. (2).

2- Find normal vector $x^2yz - 3x = 2$ at $(1,1,1)$. (5).

3)- If $\Phi\, dr = 4t^2$. Find $\int_c \Phi \cdot d\vec{r}$ from $t=0$ to $t=1$. (3).

4)- $\vec{A} = 2\hat{i} - 3\hat{k}$, $\Phi = y + xz$. Find $A \cdot \nabla\Phi$. (2).

5)- A body having mass 2 moves with $\vec{r} = (2t^2+3t)\hat{i} + t^3\hat{j} - 3t\hat{k}$ having momentum $2(4t+3)\hat{i} + 6t^2\hat{j} - 6\hat{k}$.

Find force. (3).

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mth622
1: def of kinetic energy nd formual
2:translatiin nd rotation ka formula
3:devergnce find krna ta
4:integral dia ta find krna therom ka
5:unit normal btna ta fig de v te
6:prependicular therom ki example te
7:1 ques torque waly s ta
or 1 mcqs tuff thy subjctve easy ta

Mth622 buht mushkil thy mcqs
Sub formula pochy thy moment of inertia sy bhara hoa paper tha
Hemisphere ka aya tha
Euler theorem defination
Scalar kinetic energy
M.I of small portion of hoop
Greens theorem defination
Long buht zyada easy thy masses given thy formula lgana tha
Orbital and spin angular momentum of moon aya tha
Mcqs buht buray h wy or short men b kaffi mushkil h ng asan thy

Aa

Mth622
Mcqs kafi tough thy mery
3 moment of inertia of diffrent shap
3 green theorem ma say
Short q
Kinitic energy ki definition plus formolaor translational or rotional k. E energy
Solid spher ir circular disk k formuly
Long q
Moment of conservation field ka q tha.
M. I of circular disk drive krna tha
Ak or tha jis ma angular velocity given thy simple velocity find krni thy
8:12 AM

Qurat Ul Annie
Forwarded

(A)
Q: state chalas theorm (2)
Q: define k.E nd writs its formula(2)
Q:write m.i of hollow cylider(some values are given) (2)
Q: define rotational nd translational k.E nd writs their formula (3)
Q: find values of constants using initial conditions (3)
Q: find dm of disc(daigrm given) (3)
Q:find torqe(example of handouts) 5 marks
Q: module 166 qtion 5 marks
Q: module 161 example just find
I(one) 5marks
8:12 AM

Qurat Ul Annie
Forwarded

622. Sphere hollow sphere walay saray formulas yad karnay hain obj sub doonon mai ay. Moment of inertia find krna tha. Mass m1 m2 given they v1 given v2 find krna tha moment of inertia se papr zyada tha coservative law ki necessary conditions likhi thi curvilniear motion ki examples likhni thi
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Qurat Ul Annie
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Mth622
Mcqs kafi tough thy mery
3 moment of inertia of diffrent shap
3 green theorem ma say
Short q
Kinitic energy ki definition plus formolaor translational or rotional k. E energy
Solid spher ir circular disk k formuly
Long q
Moment of conservation field ka q tha.
M. I of circular disk drive krna tha

Type a message

TODAY

Farzana Kiran

mth622
1: def of kinetic energy nd formual
2:translatiin nd rotation ka formula
3:devergnce find krna ta
4:integral dia ta find krna therom ka
5:unit normal btna ta fig de v te
6:prependicular therom ki example te
7:1 ques torque waly s ta
or 1 mcqs tuff thy subjctve easy ta

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Daud

Today mth 622. Mostly from mid.
Mcqs and long.
1long aya parellal axis theoran
1 law of conservative momentum
1 sHm sa
1 green theoran identiy proof krni.
Mcqs 5 from moment of inertia bqi mid sab
Short. Kinetic energy
Rot and translational energy
Aik curl sa tha aik divergence.

Mth622 my complete paper

1..diagram given.. Unit normal vector on surface? (2)

2..define moment of inertia and write expression for ixx and iyy (2)

3.. K. E and formlua. (2)

4..write izz M. I for z-axis. (2)

5..prove that integral(c) =A.dr=0 for every closed curve del*A=0. (3)

6..if r=acoswti+bsinwtj and p=-amwsinwti+bmwcoswtj find angular momentum? (3)

7..two masses was given and one velocity.. Find second velocity.. (3)

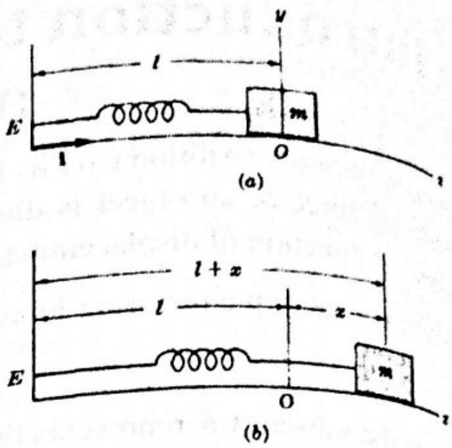
8..M.I shaded y=kx2. (3)

9...find D.E and initial conditions (5)

10..derive linear momentum formula(5)

11..solid circular cylinder of M.I. (5)

1



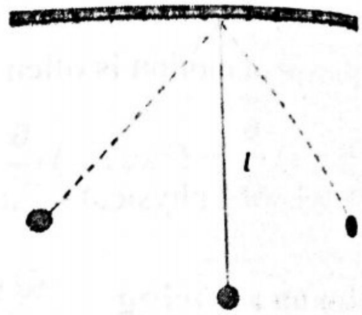
The equation representing the period for the mass attached on a spring

$$T = 2\pi\sqrt{\frac{m}{k}}$$

The above equation expresses that the time period of oscillation is independent of amplitude as well as the acceleration.

Simple pendulum

The movement of the mass attached to a simple pendulum is considered as simple harmonic motion.



The time period of a mass m attached to a pendulum of length *l* with gravitational acceleration g is given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

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It is measure in hertz (Hz).

The frequency of S.H.M can be calculate by using the following relation

$$f = \frac{1}{T}$$

Energy of S.H.M

If T is the kinetic energy, V the potential energy and $E = T + V$ the total energy of a simple harmonic oscillator, then we have

$$K.E = T = \frac{1}{2}mv^2$$

K.E = ½mv²

P.E = ½kx²

and

$$V = \frac{1}{2}kx^2$$

Then the total energy of S.H.M will be

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$y = 10\cos\ldots$

This gives the position at any time.

iii. Speed at any time?

From (6) $\frac{dy}{dt} = -10\sqrt{3/2}\sin\sqrt{3/2}t$

This gives the speed at any time.

iv. Amplitude, T, F ?

General form for the position of a particle moving to and fro is

$$y(t) = A\cos(2\pi t/T)$$

Thus, $y = 10\cos\sqrt{3/2}t$

Amplitude $= 10$

Period $= T = 2\sqrt{\frac{2}{3}}\pi$

Frequency $= 1/T$

Module No. 106

Damped Harmonic Oscillator

The forces acting on a harmonic oscillator are called damping forces which tend to decrease the amplitude of the successive oscillations or simply force apposing the motion. Since the damping force is proportional to the velocity. Thus, mathematically,

$$F_d = -bv$$

$$= -b\dot{x}$$

where d represents the damping force and b is the damping coefficient.

The negative sign shows that that direction of F_d is opposite to the velocity v.

As we know that for the restoring force:

$$m\ddot{x} + kx = 0$$

Adding the restoring force with the damping force, the equation of motion of the damped harmonic oscillator will be,

$$m\ddot{x} + kx = -b\dot{x}$$

or

$$m\ddot{x} + b\dot{x} + kx = 0$$

Since the mass is not equal to zero, therefore $\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$

Module No. 108

Euler's Theorem – Derivation

The following theorem, called Euler's theorem, is fundamental in the motion of rigid bodies.

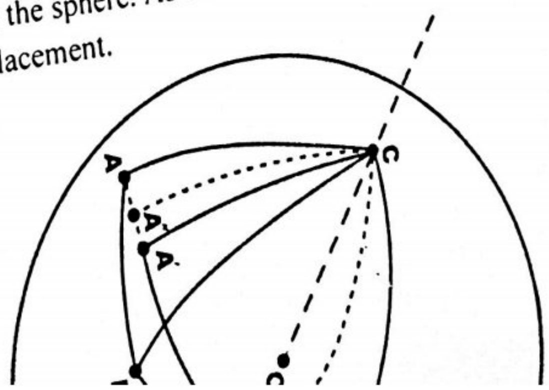
Theorem Statement

"A rotation of a rigid body about a fixed point of the body is equivalent to a rotation about a line which passes through the point."

The line referred to is called the instantaneous axis of rotation. Rotations can be considered as finite or infinitesimal. Finite rotations cannot be represented by vectors since the commutative law fails. However, infinitesimal rotations can be represented by vectors.

Proof:

Let O be the fixed point in the body, which we take as a sphere S. Further, we take O at the center of the sphere. Let A, B be two distinct points on the sphere. As the body moves, the point O (on the axis) remains foxed and A and B suffer displacement.



Module No. 109

Chasles' Theorem

Theorem Statement:

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line (called its screw axis) followed (or preceded) by a rotation about that line.

Explanation:

- A rigid body has six degrees of freedom.
- By Euler's theorem, three of these are associated with pure rotation.
- The remaining three must be associated with translation.
- To describe the general motion of a rigid body, think of the general motion as translation of a fixed point O in the body to a point O' followed by the rotation about an axis through O'.

Module No. 110

Kinematics of a System of Particles (Space, time & matter)

Kinematics is the branch of mechanics deals with the moving objects without reference to the forces which cause the motion.
In other words we can say those kinematics are the features or properties of motion of concerned with system of particles (rigid bodies).
Here some features of rigid body motion are

- Displacement
- Position
- Velocity
- Linear Velocity & Angular Velocity
- Linear Acceleration & Angular Acceleration
- Motion of a Rigid Body (Translation & Rotation)

Module No. 111

The Concept of Rectilinear Motion of particles, Uniform Rectilinear Motion, Uniformly Accelerated Rectilinear Motion

When a moving particle remains on a single straight line, the motion is said to be rectilinear. In this case, without loss of generality we can choose the x-axis as the line of motion.
The general equation of motion is then

$$F = ma \Rightarrow F(x, \dot{x}, \ddot{x}) = m\ddot{x}$$

Rectilinear motion for a particle:

$x(t)$

Rectilinear motion of a body is defined by considering the two point of a body covered the same ... the parallel direction. The figures below illustrate rectilinear motion for a particle and

Module No. 112

The Concept of Curvilinear Motion of Particles

The motion of a particle moving in a curved path is called curvilinear motion. Example: A stone thrown into the air at an angle.

Curvilinear motion describes the motion of a moving particle that conforms to a known or fixed curve. The study of such motion involves the use of two co-ordinate systems, the first being planar motion and the latter being cylindrical motion.

Tangential and normal unit vectors are usually denoted by $\overrightarrow{e_t}$ and $\overrightarrow{e_n}$ respectively.

Velocity of Curvilinear motion

If the tangential and normal unit vectors are $\overrightarrow{e_t}$ and $\overrightarrow{e_n}$ respectively, then the velocity will be

$$\vec{v} = \frac{ds}{dt}\overrightarrow{e_t}$$

You have already learnt that

$$v = \mathrm{v}\mathrm{T}$$

Acceleration of Curvilinear motion

If the tangential and normal unit vectors are $\overrightarrow{e_t}$ and $\overrightarrow{e_n}$ respectively, then the acceleration will be

$$\vec{a} = \frac{d^2s}{dt^2}\overrightarrow{e_t} + \frac{\left(ds/_{dt}\right)^2}{\rho}\overrightarrow{e_n}$$

You have already learnt that

$$a = \mathrm{T}\frac{d\mathrm{v}}{dt} + \frac{\mathrm{v}^2}{\mathrm{r}}\mathrm{N}$$

Example

- A stone thrown into the air at an angle.
- A car driving along a curved road.
- Throwing paper airplanes or paper darts is an example of curvilinear motion.

Example 2

Problem Statement

Prove

$$\frac{de_\rho}{dt} = \varphi \dot{e}_\varphi$$
$$\frac{de_\varphi}{dt} = -\dot{\varphi} e_\rho$$

where dots denote differentiation with respect to time t.

Solution

We have

$$e_\rho = \cos\varphi\,\hat{\imath} + \sin\varphi\,\hat{\jmath}$$

and

$$e_\varphi = -\sin\varphi\,\hat{\imath} + \cos\varphi\,\hat{\jmath}$$

Then

$$\frac{de_\rho}{dt} = \frac{d}{dt}(\cos\varphi\,\hat{\imath} + \sin\varphi\,\hat{\jmath})$$
$$= -\sin\varphi\,\dot{\varphi}\hat{\imath} + \cos\varphi\,\dot{\varphi}\hat{\jmath} = \dot{\varphi}(-\sin\varphi\,\hat{\imath} + \cos\varphi\,\hat{\jmath}) = \dot{\varphi} e_\varphi$$
$$\frac{de_\varphi}{dt} = \frac{d}{dt}(-\sin\varphi\,\hat{\imath} + \cos\varphi\,\hat{\jmath})$$
$$= -\cos\varphi\,\dot{\varphi}\hat{\imath} - \sin\varphi\,\dot{\varphi}\hat{\jmath} = -\dot{\varphi}(\cos\varphi\,\dot{\varphi}\hat{\imath} + \sin\varphi\,\dot{\varphi}\hat{\jmath}) = -\dot{\varphi} e_\rho$$

Module No. 115

Conservation of Energy for a System of Particles

Statement

"The law of conservation of energy describes that the net energy of an isolated system remains conserved. Energy can neither be created nor destroyed; rather, it transforms from one form to another."

Theorem Statement
(Principle of conservation of energy)

In case of conservative force field, the total energy is a constant. i.e.,

If T is for kinetic energy and V is for potential energy, then the total energy E is

$E = T + V = \text{constant}$

Explanation

For a conservative force field, we have already learned that the work done by the system of particles is

$$W = T_2 - T_1$$

Module No. 118

Example of Impulse

Well

Example:
What is the magnitude of the impulse developed by a mass of 200 gm which changes its velocity from $5i - 3j + 7k$ m/sec to $2i + 3j + k$ m/sec?

Solution:
Since we have the following information:

$$m = 200 \text{ gm}$$
$$v_1 = 5i - 3j + 7k$$
$$v_2 = 2i + 3j + k$$

As we know that

$$\text{Impulse} = p_2 - p_1$$
$$= mv_2 - mv_1$$
$$= m(v_2 - v_1)$$

Substituting the values, we get

$$\text{Impulse} = 200(2i + 3j + k - (5i - 3j + 7k))$$
$$= 200(-3i + 6j - 6k)$$

The magnitude of the Impulse will be

$$\text{Impulse magnitude} = 200\sqrt{9 + 36 + 36}$$
$$= 200\sqrt{81}$$
$$= 200(9)$$
$$= 1800 \text{ gm/sec}$$
$$= 1.8 \text{ N sec}$$

Module No. 119

Torque

Definition

Torque is defined as the turning effect of a body. It is trend of an acting force due to which the rotational motion of a body changes. It is also called twist and rotational force on an object. Mathematically, torque is defined as the cross product of the force vector to the distance vector which causes rotational motion of the body.

$$\tau = \vec{r} \times \vec{F}$$

The magnitude of torque depends upon the applied force, the length of the lever arm connecting the axis to the point where the force applied, and the angle between the force vector and the length of lever arm. Symbolically we can write it as:

$$\tau = |r||F| \sin \theta$$

Torque is a vector quantity implies that it has direction as well as magnitude.
The SI unit for torque is the newton meter (Nm).
The direction of torque can be approximate using **Right Hand Rule.**

Theorem:

The torque acting on a particle equals the time rate of change in its angular momentum. i.e.,

$$\tau = \frac{d\Omega}{dt}$$

where $\Omega = r \times p$ is defined angular momentum of the system.

Proof:

As we know torque is defined as

$$\tau = r \times F = r \times ma$$

as m is constant, therefore

$$= m(r \times a)$$

$$= m(r \times \frac{dv}{dt})$$

$$= m\frac{d}{dt}(r \times v)$$

$$= \frac{d}{dt}(r \times mv)$$

$$= \frac{d}{dt}(r \times p)$$

where $p = mv$ is defined as linear momentum and hence the theorem.

Module No-120

Example of Torque

Example

A particle of mass m moves Along a space curve defined by $r = a\cos\omega t\, i + b\sin\omega t\, j$.

Find

(i) The Torque

(ii) The angular momentum about the origin.

Solution

(i) As we know that the torque acting on a particle is equal to the time rate of change of its angular momentum. i.e.

$$\tau = \frac{d\Omega}{dt} = \frac{d}{dt}(r \times p)$$

where $p = mv$.

As $r = a\cos\omega t\ i + b\sin\omega t\ j$. Therefore

$$v = \frac{dr}{dt}$$

$$v = -a\omega\sin\omega t\, i + b\omega\cos\omega t\, j$$

So.

$$p = -am\omega\sin\omega t\, i + bm\omega\cos\omega t\, j$$

Now

$$r \times p$$

$$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a\cos\omega t & b\sin\omega t & 0 \\ -am\omega\sin\omega t & bm\omega\cos\omega t & 0 \end{vmatrix}$$

$$= [abm\omega\cos^2\omega t + abm\omega\sin^2\omega t]\hat{k}$$

$$= abm\omega\hat{k}$$

Module No. 122

Properties of Rigid Bodies

Following are some of the properties of the rigid bodies.

Degree of freedom

The number of coordinates required to specify the position of a system of one or more par called the number of degrees of freedom of the system.
For example a particle moving freely in space requires 3 coordinates, e.g. (x, y, z), to spe position. Thus the number of degrees of freedom is 3.
Similarly, a system consisting of N particles moving freely in space requires 3N coordir specify its position. Thus the number of degrees of freedom is 3N.

Translations

A displacement of a rigid body is a direct change of position of its particles. Translational m is the displacement of all particles of the body by the same amount and the line segment j the initial and the final position of the particles represented by parallel vectors.
Examples of translational motion are particles freely falling down to earth and the motio bullet fired from a gun.

Rotations

Circular motion of a body about a fixed point or axis is called rotation. If during a displacem the points of the rigid body on some line remains fixed and all other are displaced through th same angle, then this displacement is called rotation. A rigid performs rotations around an imaginary line called a rotation axis.
If the axis of rotation passes through the center of mass of the rigid body then body is said to spin or rotate upon itself. If a body rotates about some external fixed point is called revolutio orbital motion of the rigid body. The example of revolution is the rotation of earth around sun and motion of moon around sun.
Rotational motion concerns only with rigid bodies. The reverse rotation of a body (inverse rotation) is also a rotation.
A wheel is common examples of rotation.

Module No. 126

Example of moment of Inertia and Product of Inertia

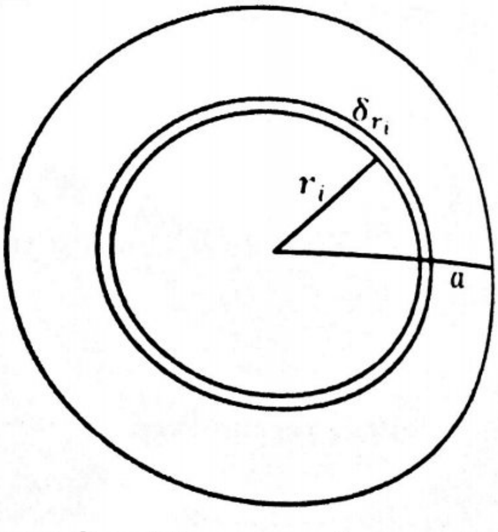
Problem Statement:

Find the moments of inertia of a ring of radius a about an axis through center.

Solution:

Let M be the mass and a the radius of the ring. Then the mass per unit length will be $M/2\pi a$. We regard this ring to be composed of small elements of mass (δm) each of length δs, we can write it as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi a} \Rightarrow \delta m = \frac{M}{2\pi a}\delta s$$



Moment of inertia of the element about an axis through center O and the perpendicular to the plane of the ring equals $\frac{M}{2\pi a}\delta s\, a^2$.

Therefore the M.I of the whole ring will be

$$I_{ring} = \frac{M}{2\pi a}a^2 \sum_{elements} \delta s = \frac{Ma}{2\pi}\int ds = \frac{Ma}{2\pi} \times 2\pi a = Ma^2$$

Hence Ma^2 is the required moment of inertia of the ring.

Module No. 132
Example of Conservation of Momentum

Problem Statement

A particle A, of mass 6 kg, travelling in a straight line at 5 ms^{-1} collides with a particle B, of mass 4 kg, travelling in the same straight line, but in the opposite direction, with a speed of 3 ms^{-1}. Given that after the collision particle A continues to move in the same direction at 1.5 ms^{-1}, what speed does particle B move with after the collision? (© mathcentre 2009)

Given Data

Mass of particle $A = m_1 = 6\text{kg}$
Velocity of A before collision $= u_1 = 5\text{ms}^{-1}$
Velocity of A after collision $= v_1 = 1.5\text{ms}^{-1}$
Mass of Particle $B = m_2 = 4\text{kg}$
Velocity of B before collision $= u_2 = 3\text{ms}^{-1}$

Required

Velocity of B after collision $= v_2 = ?$

Solution

It is always useful to depict the collision with the velocities both before and after.
Using the principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$
$$6 \times 5 + 4 \times (-3) = 6 \times 1.5 + 4 \times v_2$$
$$9 = 4 \times v_2$$
$$v_2 = 9/4 = 2.3\, m\, s^{-1}$$

So after the collision particle B moves with a speed of 2.3 m s−1, in the same direction as A.

Example 2

Module No. 143
Introduction to Special Moments of Inertia

In this module, we will study some special moment of inertia. In order to introduce the moment of inertia of some specific shapes and special rigid bodies, we assume that the rigid body is of uniform density of particles.

Solid Circular Cylinder

We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder.

$$I = \frac{1}{2} M a^2$$

Hollow Circular Cylinder

We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder.
We consider the thickness of Wall of cylinder is negligible.

$$I = M a^2$$

Solid Sphere

We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter.

$$I = \frac{2}{5} M a^2$$

Hollow Sphere

We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter.
We consider the thickness of sphere is negligible.

$$I = M a^2$$

Rectangular Plate

We consider sides of length $\boldsymbol{a}$ and $\boldsymbol{b}$. and mass M about an axis perpendicular to the plate through the center of mass.

$$I = \frac{1}{12} M(a^2 + b^2)$$

Thin Rod

We assume the length of rod is $\boldsymbol{a}$ and mass M about an axis perpendicular to the rod through the center of mass.

$$I = \frac{1}{12} Ma^2$$

Triangular Lamina

We assume the height h and mass **M** of lamina.

$$I = \frac{1}{6} Mh^2$$

Right Circular Cone

We assume the radius of the circular cone is a and mass M.

$$I = \frac{3}{10} Ma^2$$

Thin Road $= \frac{1}{3} Ma^2$

Hoop-Ring $= Mr^2$

Annular Disk $= \frac{1}{2} M(R_2^2 + R_1^2)$

Circular Disk $= \frac{1}{2} Ma^2$

Rectangular plate $= \frac{1}{3} Ma^2$

Sequare plate $= \frac{Ma^2}{12}$

Triangular Lamina $= \frac{1}{6} Mh^2$

Elliptical Plate $= \frac{M}{\pi ab}$

Solid circular cylinder $= \frac{1}{2} Ma^2$

Hollow Shell $= MR^2$

Solid Sphere $= \frac{2}{5} Ma^2$

Hollow Sphere $= \frac{2}{3} MR^2$

Module No. 144

M.I. of the Thin Rod – Derivation

Problem Statement

Calculate the moment of inertia of a uniform rod of length l about an axis perpendicular to the rod and passing through an end point.

Solution

Let the $X-axis$ be chosen along the length of the rod, with origin at one end point as shown in the figure. Let M and a be the mass and length of rod respectively. We suppose the rod to be composed of small elements.

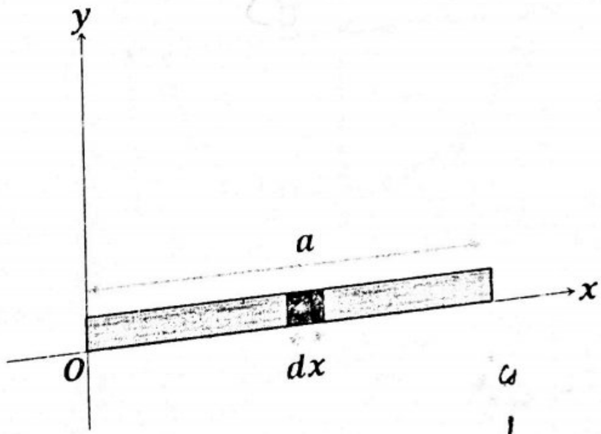
Let dm and dx be the mass and the length of the specific element of the rod at a distance x from the end point O.

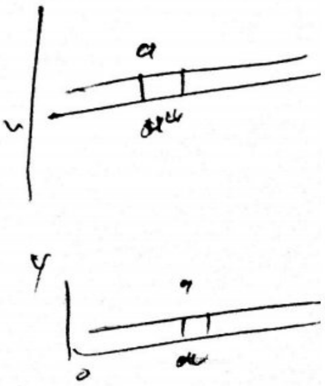
$$\frac{dm}{du} = \frac{M}{a}$$

$$dm = \frac{m}{a}(dx)$$

$$dmu^2 = \int_0^a \frac{m}{a} u^2 \, du$$

$$= \frac{m}{a}\left(\frac{a}{3} - 0\right)$$





Then
$$\frac{dm}{dx} = \frac{M}{a}$$

$$\Rightarrow dm = \frac{M}{a}dx$$

Then the moment of inertia of this element about the given axis is

$$I_{element} = dmx^2 = \frac{M}{a}x^2dx$$

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Hence the moment of inertia of the whole rod will be

$$I = \sum_{all\,elements} \frac{M}{a} x^2 dx$$

$$= \frac{M}{a} \int_0^a x^2 dx$$

$$= \frac{M}{a} \cdot \left| \frac{x^3}{3} \right|_0^a = \frac{M}{a} \frac{a^3}{3}$$

$$= \frac{1}{3} M a^2$$

85

v.i.p

Module No. 145

M.I. of Hoop or Circular Ring – Derivation

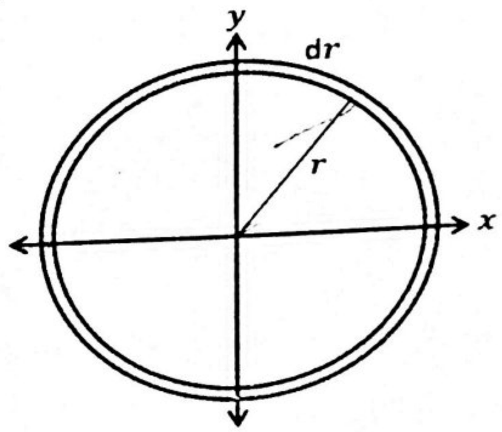
Problem Statement

Calculate the moment of inertia of a hoop of mass M and radius r about an axis passing through its center.

Proof

Let M be the mass and r the radius of the hoop. Then we can define the density of the hoop by

$$\rho = \frac{\text{mass}}{\text{area}} = \frac{M}{2\pi r dr}$$



We consider this hoop to be composed of small masses (δm) each of length δs.
We can write it as

$$\rho = \frac{M}{2\pi r dr} = \frac{\delta m}{dr\delta s}$$ $\delta m = \frac{m\,\delta s}{2\pi r}$

$$\Rightarrow \delta m = \frac{M}{2\pi r}\delta s$$

Moment of inertia of the small portion of the hoop of mass δm about an axis through center and perpendicular to the plane of the ring equals

$$I_{particle} = \delta m r^2$$

$$= \frac{M}{2\pi r}\delta s\, r^2 = \frac{Mr}{2\pi}\delta s$$

Therefore the $M.I$ of the whole ring/hoop will be

$$I = \frac{Mr}{2\pi}\sum_{particles} \delta s$$

$$I = \frac{Mr}{2\pi}\int ds$$

$$= \frac{Mr}{2\pi} 2\pi r$$

$$= Mr^2$$

- **Hence we obtain Mr^2 as the moment of inertia of the hoop.**

Module No. 149

M.I. of Square Plate – Derivation

Problem Statement

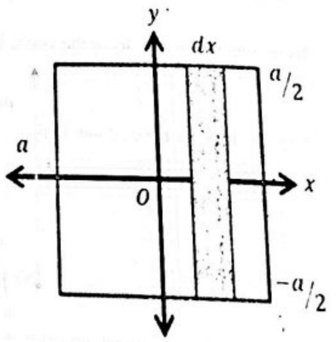
Calculate the moment of inertia of a uniform square plate with sides a about any axis through its center and lying in the plane of plate.

Solution

Consider a uniform square plate with length of its side to be a. we have to find out M.I. of this plate about any axis, through its center.
We consider this axis as $y - axis$.
Let ρ be the density of plate and the total area of square plate is a^2.
So density will be

$$\rho = \frac{M}{a^2}$$

We assume that the plate has been divided into vertical strips. Let us consider a strip from the whole square as shown in figure.



The strips are chosen in this way because each point on a particular strip is approximately the same distance from axis of rotation i.e. y-axis, the mass of the strip is δm and the width of each strip is δx, then the area of the strip will be $a\delta x$, so

$$\delta m = \rho a \delta x$$

Let the distance of the strip from y-axis is x, then the moment of inertia of strip will be

$$\delta I = \delta m x^2 = x^2 \rho a \delta x$$

So the moment of inertia of square is

$$I_{square} = \sum_{all\ strips} x^2 \rho a \delta x$$

$$I_{square} = \int_{-a/2}^{a/2} x^2 \rho a dx = \rho a \int_{-a/2}^{a/2} x^2 dx$$

$$= \rho a \left|\frac{x^3}{3}\right|_{-a/2}^{a/2} = \frac{\rho a}{3}\left|\frac{a^3}{8} - \frac{(-a^3)}{8}\right|$$

$$= \frac{\rho a}{3}\left(\frac{a^3}{4}\right) = \frac{\rho a^4}{12}$$

we substitute $\rho = \frac{M}{a^2}$, we obtain

$$I_{square} = \frac{Ma^2}{12}$$

Module No. 151

M.I. of Elliptical Plate along its Major Axis – Derivation

Problem Statement

Find the M.I. of a uniform elliptical plate with semi major axes and semi minor axes a, b respectively about its major axes.

Solution

We consider the elliptical plate as (1)

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

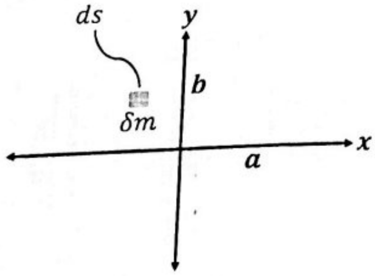
with semi major axes along x-axis as shown in figure.

From (1) we have

$$y = \pm \frac{b}{a}\sqrt{a^2 - x^2}$$

So, let $y_1 = \frac{b}{a}\sqrt{a^2 - x^2}$.

We consider a small element of plate of mass δm of the elliptical plate will be ρds which will be equal to $\rho dxdy$. The moment of inertia if this element along x-axis will be equal to $I = \delta m y^2$.



Then the moment of inertia of whole plate will be

$$I_x = \int (\delta m) y^2 = \int_{plate} \rho ds\, y^2$$

$$= \rho \int_{-a}^{a} \left(\int_{-\frac{b}{a}\sqrt{a^2-x^2}}^{\frac{b}{a}\sqrt{a^2-x^2}} y^2 dy \right) dx$$

$$= \rho \int_{-a}^{a} \frac{2y_1^2}{3} dx = \frac{2\rho}{3} \int_{-a}^{a} \frac{b^3}{a^3}(a^2 - x^2)^{3/2} dx$$

...symmetry, we can write it as

$$= \frac{4\rho b^3}{3a^3} \int_0^a (a^2 - x^2)^{3/2} dx$$

...making use of polar coordinates, substitute $x = a\sin\theta$, then $dx = a\cos\theta$ this integral becomes

$$I_x = \frac{4\rho b^3}{3a^3} \int_0^{\pi/2} a^3 \cos^3\theta\, (a\cos\theta) d\theta$$

$$= \frac{4\rho a b^3}{3} \int_0^{\pi/2} \cos^4\theta\, d\theta$$

$$= \frac{4\rho a b^3}{3} \frac{1 \times 3}{2 \times 4} \times \frac{\pi}{2}$$

$$= \frac{1}{4} \rho a b^3 \pi$$

$$= \frac{1}{4} a b^3 \pi \times \frac{M}{\pi a b} = \frac{1}{4} M b^2$$

...where for elliptical plate, we have

$$\rho = \frac{M}{\pi a b}$$

...the required expression for M.I.

Module No. 156

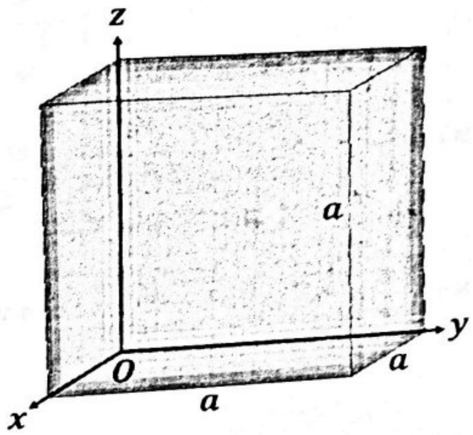
Inertia Matrix / Tensor of solid Cuboid

Problem Statement

Calculate the inertia matrix / inertia tensor of uniform solid cube at one of its corners.

Solution

Let the length of the edges be a of each side and let the axes be chosenalong the edges as shown in the figure.



By definition

$$I_{11} \equiv I_{xx} = \int \rho(r)(y^2 + z^2)dV$$

Since the box is made of uniform material, the density ρ must be constant. Therefore

$$I_{xx} = \rho \int_0^a \int_0^a \int_0^a (y^2 + z^2)dxdydz$$

$$= \rho \int_0^a \int_0^a (y^2 + z^2)dydz \int_0^a dx$$

$$= \rho a \int_0^a \int_0^a (y^2 + z^2)dydz$$

$$= \rho a\left[\int_0^a\int_0^a y^2 dydz + \int_0^a\int_0^a z^2 dydz\right]$$

$$= \rho\left(\frac{a^4}{3} + \frac{a^4}{3}\right)$$

$$I_{xx} = \rho a\left(\frac{2a^4}{3}\right) = \rho\left(\frac{2a^5}{3}\right)$$

Since we know that for the cube

$$\rho = \frac{M}{a^3}$$

We obtain

$$I_{xx} = \frac{2M}{3}a^2$$

Similarly, due to symmetry, we can write

$$I_{yy} = \frac{2M}{3}a^2$$

$$I_{zz} = \frac{2M}{3}a^2$$

Now for the product of inertia, we have

$$I_{12} = -\int \rho xydV$$

$$= -\rho\int_0^a\int_0^a\int_0^a xydxdydz$$

$$= -\rho\int_0^a xdx\int_0^a ydy\int_0^a dz$$

$$= -\rho a\frac{a^2}{2}\frac{a^2}{2} = -\rho\frac{a^5}{4}$$

Again using

$$\rho = \frac{M}{a^3}$$

We obtain

$$I_{12} = -\frac{Ma^2}{4}$$

Similarly,

$$I_{12} = -\frac{Ma^2}{4}$$

$$I_{12} = -\frac{Ma^2}{4}$$

The required inertia matrix / inertia tensor will be

$$I_{ij} = \begin{bmatrix} \frac{2M}{3}a^2 & -\frac{Ma^2}{4} & -\frac{Ma^2}{4} \\ -\frac{Ma^2}{4} & \frac{2M}{3}a^2 & -\frac{Ma^2}{4} \\ -\frac{Ma^2}{4} & -\frac{Ma^2}{4} & \frac{2M}{3}a^2 \end{bmatrix}$$

where ω is the angular velocity of the system when the boy

$$\omega = \frac{120(5) + 0}{120 + 120}$$
$$= 0.75 \text{ radian per sec}$$

Module No. 161

Example 2 of Moment of Inertia

Problem Statement

A boy of mass $M = 30\text{kg}$ is running with a velocity of $3\ \text{m/sec}$ on ground just tangentially to a merry-go-round which is at rest. The boy suddenly jumps on the merry-go-round. Calculate the angular velocity acquired by the system. The merry-go-round has a radius of $r = 2\text{m}$ and a mass $m = 120\text{kg}$ and its moment of inertia is $120\ \text{kgm}^{-2}$.

Solution

The merry-go-round rotates about an axis which we regard as passing through its c.m.

Let's give the following notations:

- M.I of boy $= I_1$
- M.I of merry go round $= I_2$
- Vel. of boy $= v_1$
- Vel. of merry go round $= v_2$
- Angular vel. of boy $= \omega_1$
- Angular vel. of merry go round $= \omega_2$

The moment of inertia I_1 of the boy about the axis of rotation can be found by

$$I_1 = Md^2 = 30 \times 2^2 = 120kgm^{-2}$$

The moment of inertia I_2 of merry go round is given to be

$$I_2 = 120kgm^{-2}$$

Since the velocity of the boy about the merry-go-round is $v_1 = 3\text{ms}^{-1}$, therefore his angular velocity ω_1 about the axis of rotation is therefore

$$\omega_1 = \frac{v}{d} = \frac{3}{2} = 1.5 \text{ radian per second}$$

Initially the merry go round is at rest, therefore $v_2 = 0$, and thus its angular velocity $\omega_2 = 0$. We ignore friction and therefore there is no external torque on the system.
Hence by the law of conservation of angular momentum

$$(I_1 + I_2)\omega = I_1\omega_1 + I_2\omega_2$$

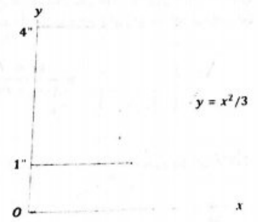
$$\omega = \frac{I_1\omega_1 + I_2\omega_2}{(I_1 + I_2)}$$

Module No. 163

Example 4 of Moment of Inertia

Problem (© engineering.unl.edu)

Calculate the moment of inertia of the shaded area given in figure about y-axis.



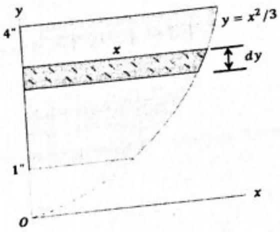
Solution

Here, we have

$$y = x^2/3 \tag{1}$$

We consider an elementary strip from the shaded area whose M.I is

$$I_{strip} = \frac{1}{3}x^2(xdy) = \frac{1}{3}x^3dy$$



Then by integration, we obtain the moment of inertia of the whole shaded area

$$I_y = \sqrt{3}\int_1^4 y^{3/2}dy = \sqrt{3}\,\frac{2}{5}\left|y^{5/2}\right|_1^4$$

$$= \sqrt{3}\,\frac{2}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}(32 - 1)$$

$$= 21.5 \text{ inch}^4$$

principal moments and directions (axes) of inertia.

Related Theorems

Theorem 1

A 3×3 symmetrical matrix has three real eigenvalues, which may be distinct or repeated.

Theorem 2

The eigenvectors of a symmetrical matrix corresponding to distinct eigenvalues are orthogonal.

Theorem 3

It is always possible to find three mutually orthogonal eigenvectors for a 3×3 symmetric matrix, whether the eigenvalues are distinct or repeated.

Results

In the light of above theorems, we deduce the following results about the inertia matrix I_{mat} at point O of a rigid body.

i. The principal of moment of inertia are always real numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where m_i and d_i are both real.

Module No. 181

Example 2 of Equation of Motion in Rotating Frame of Reference

Problem Statement

v.1

A coordinate system OXYZ rotates with angular velocity $\vec{\omega} = 2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}$ relative to the fixed coordinate system $OX_0Y_0Z_0$ both systems having the same origin. If $\vec{A} = \sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k}$ where $\hat{\imath}, \hat{\jmath}, \hat{k}$ refer to the rotating coordinate system OXYZ then find $\frac{dA}{dt}$ and $\frac{d^2A}{dt^2}$ w.r.to

i. Fixed system

ii. The rotating system

Solution

i. To find $\frac{dA}{dt}$ and $\frac{d^2A}{dt^2}$ in the fixed system, we have to apply the following formulas

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \vec{\omega} \times \vec{A} \tag{1}$$

$$\left(\frac{d^2\vec{A}}{dt^2}\right)_f = \left(\frac{d^2\vec{A}}{dt^2}\right)_r + 2\vec{\omega} \times \frac{d\vec{A}}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{A} + \vec{\omega} \times (\vec{\omega} \times \vec{A}) \tag{2}$$

$$\left(\frac{d\vec{A}}{dt}\right)_r = \left[\frac{d\vec{A}}{dt}\right]_r = \frac{d\vec{A}_r}{dt}$$

$$= \frac{d}{dt}(\sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k})$$

$$= \cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k}$$

$$\vec{\omega} \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -3 & 5 \\ \sin t & -\cos t & e^{-t} \end{vmatrix}$$

$$\vec{\omega} \times \vec{A} = (-3e^{-t} + 5\cos t)\hat{\imath} - (2e^{-t} - 5\sin t)\hat{\jmath} + (-2\cos t + 3\sin t)\hat{k}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = \cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k} + (-3e^{-t} + 5\cos t)\hat{\imath} - (2e^{-t} - 5\sin t)\hat{\jmath}$$

$$+ (-2\cos t + 3\sin t)\hat{k}$$

$$= (\cos t + 5\cos t - 3e^{-t})\hat{\imath} + (5\sin t + \sin t - 2e^{-t})\hat{\jmath} + (-e^{-t} - 2\cos t + 3\sin t)\hat{k}$$

$$= (6\cos t - 3e^{-t})\hat{\imath} + (6\sin t - 2e^{-t})\hat{\jmath} + (-e^{-t} - 2\cos t + 3\sin t)\hat{k}$$

Now for solving equation (2), we proceed as follows

$$\left(\frac{d^2\vec{A}}{dt^2}\right)_r = \frac{d}{dt}\left(\frac{d\vec{A}}{dt}\right)_r = \frac{d}{dt}(\cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k})$$

$$= (-\sin t\,\hat{\imath} + \cos t\,\hat{\jmath} + e^{-t}\hat{k})$$

$$2\vec{\omega} \times \left(\frac{d\vec{A}}{dt}\right)_r = 2(2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}) \times (\cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k})$$

$$= (4\hat{\imath} - 6\hat{\jmath} + 10\hat{k}) \times (\cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k})$$

$$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 4 & -6 & 10 \\ \cos t & \sin t & -e^{-t} \end{vmatrix}$$

$$2\vec{\omega} \times \left(\frac{d\vec{A}}{dt}\right)_r$$

$$= (6e^{-t} - 10\sin t)\hat{\imath} + (4e^{-t} + 10\cos t)\hat{\jmath} + (4\sin t + 6\cos t)\hat{k}$$

$$\vec{\omega} = 2\hat{\imath} - 3\hat{\jmath} + 5\hat{k} \Rightarrow \dot{\vec{\omega}} = \frac{d\vec{\omega}}{dt} = 0$$

So,

$$\frac{d\vec{\omega}}{dt} \times \vec{A} = 0 \times (\sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k}) = 0$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{A}) = (2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}) \times [(2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}) \times (\sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k})]$$

$$= (2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}) \times [(-3e^{-t} + 5\cos t)\hat{\imath} - (2e^{-t} - 5\sin t)\hat{\jmath} + (-2\cos t + 3\sin t)\hat{k}]$$

$$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -3 & 5 \\ -3e^{-t} + 5\cos t & -2e^{-t} + 5\sin t & -2\cos t + 3\sin t \end{vmatrix}$$

$$= (6\cos t - 9\sin t - 25\sin t + 10e^{-t})\hat{\imath} + (4\cos t - 6\sin t - 15e^{-t} + 25\cos t)\hat{\jmath} + (4e^{-t} - 10\sin t + 9e^{-t} - 15\cos t)\hat{k}$$

$$= (6\cos t - 34\sin t + 10e^{-t})\hat{\imath} + (29\cos t - 6\sin t - 15e^{-t})\hat{\jmath} + (-4e^{-t} + 10\sin t + 9e^{-t} - 15\cos t)\hat{k}$$

$$= (-\sin t\,\hat{\imath} + \cos t\,\hat{\jmath} + e^{-t}\hat{k}) + (6e^{-t} - 10\sin t)\hat{\imath} + (4e^{-t} + 10\cos t)\hat{\jmath} + (4\sin t + 6\cos t)\hat{k} + 0 + (6\cos t - 34\sin t + 10e^{-t})\hat{\imath} + (29\cos t - 6\sin t - 15e^{-t})\hat{\jmath} + (19\sin t - 13e^{-t} + 15\cos t)\hat{k}$$

$$\left(\frac{d^2\vec{A}}{dt^2}\right)_I = (6\cos t - 45\sin t + 16e^{-t})\hat{\imath} + (40\cos t - 6\sin t - 11e^{-t})\hat{\jmath} + (14\sin t - 12e^{-t} + 21\cos t)\hat{k}$$

ii. The rotating system

$$\left(\frac{d\vec{A}}{dt}\right)_r = \left|\frac{d\vec{A}}{dt}\right|_r = \frac{d\vec{A}_r}{dt}$$

$$= \frac{d}{dt}(\sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k})$$

$$= \cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k}$$

$$\left(\frac{d^2\vec{A}}{dt^2}\right)_r = \frac{d}{dt}\left(\frac{d\vec{A}}{dt}\right)_r$$

$$= \frac{d}{dt}(\cos t\,\hat{\imath} + \sin t\,\hat{\jmath} - e^{-t}\hat{k})$$

$$= (-\sin t\,\hat{\imath} + \cos t\,\hat{\jmath} + e^{-t}\hat{k})$$